

Data Visualization



Introduction of Visualization



“The use of computer-supported, interactive, visual representations of abstract data to amplify cognition”

Card, Mackinlay, & Shneiderman

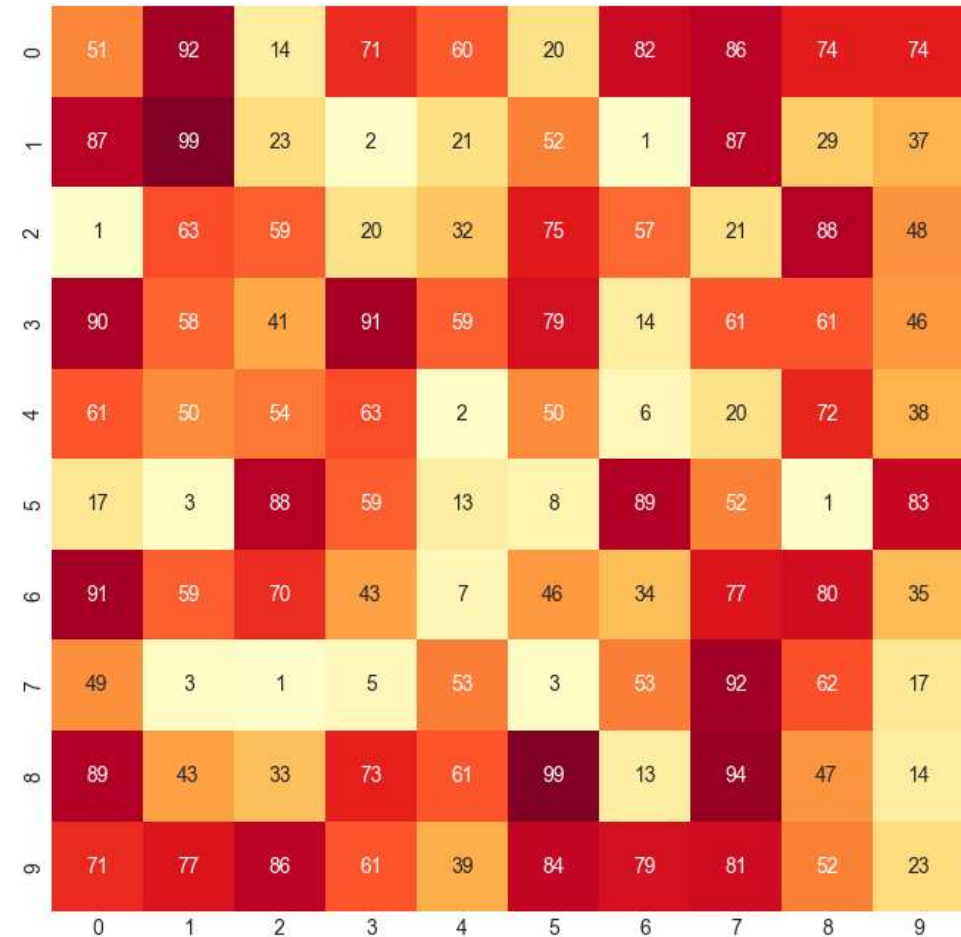
- **Computer-supported:** Not hand-drawn.
- **Interactive:** Zoom, filter, hover.
- **Abstract Data:** Stocks, Votes, Sales (*Non-physical data*)
- **Amplify Cognition:** Make you smarter.

Introduction of Visualization



51	92	14	71	60	20	82	86	74	74
87	99	23	2	21	52	1	87	29	37
1	63	59	20	32	75	57	21	88	48
90	58	41	91	59	79	14	61	61	46
61	50	54	63	2	50	6	20	72	38
17	3	88	59	13	8	89	52	1	83
91	59	70	43	7	46	34	77	80	35
49	3	1	5	53	3	53	92	62	17
89	43	33	73	61	99	13	94	47	14
71	77	86	61	39	84	79	81	52	23

Raw Data



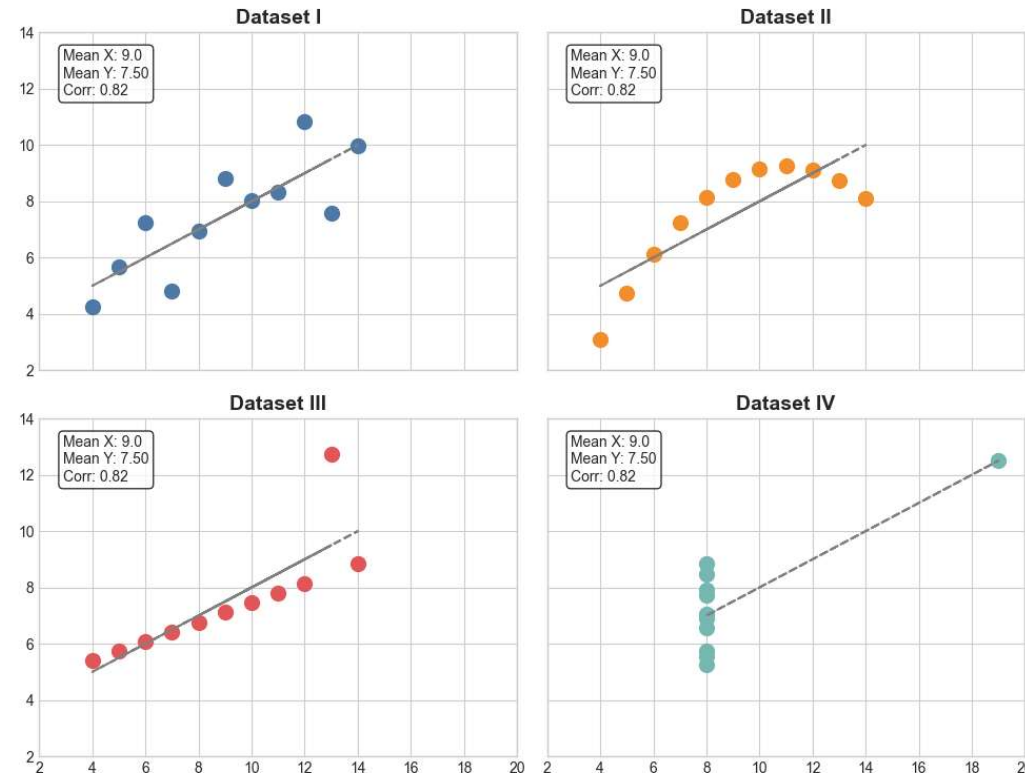
Heatmap

Introduction of Visualization

Anscombe's Quartet Data Set

X_1	Y_1	X_2	Y_2	X_3	Y_3	X_4	Y_4
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89
mean : 9	7.50	9	7.50	9	7.50	9	7.50
Variance : 11	4.12	11	4.12	11	4.12	11	4.12

Anscombe's Quartet: Same Stats, Different Stories



Data Abstraction



Data abstraction



- ***Data level***

- There are five basic **data types**
 - *Attribute* – property that can be measured
 - Salary, price, temperature,...
 - *Item* – individual entity
 - A row of a table, a node in a network...
 - *Link* – relationship between items
 - *Position* – spatial/ location
 - Latitude–longitude,
 - *Grid* – how to samples continuous data
 - Geometric, topological

- Data semantics

- ***Dataset level***

- There are four basic types
 - Tables
 - Networks
 - Fields
 - Geometry

Data level: Items & Attributes

- **item**: individual entity, discrete
 - eg patient, car, stock, city
 - *"independent variable"*

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Data level: Items & Attributes

- **item**: individual entity, discrete
 - eg patient, car, stock, city
 - "*independent variable*"
- **attribute**: property that is measured, observed, logged...
 - eg height, blood pressure for patient
 - "*dependent variable*"

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Dataset level

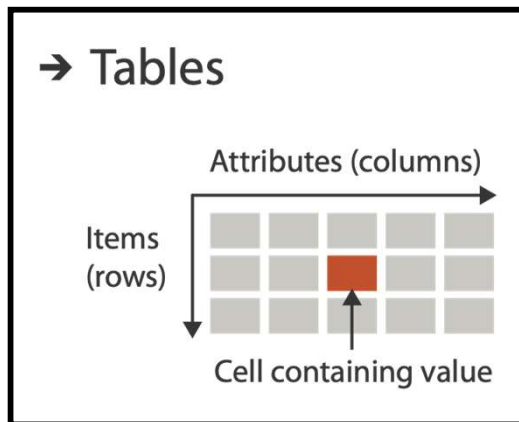
Tables

Items

Attributes

- **flat table**
 - one item per row
 - each column is attribute
 - cell holds value for item-attribute pair

attributes: name, age, shirt size, fave fruit



Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

Dataset level

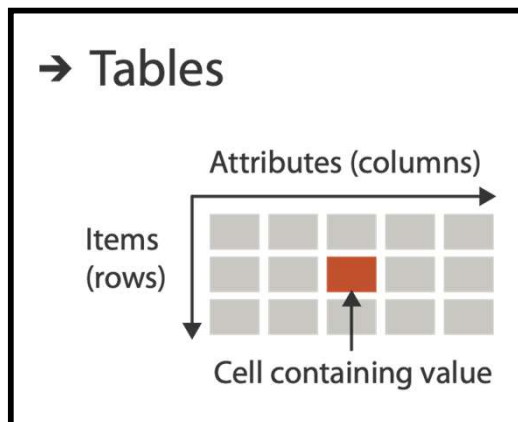
Tables

Items

Attributes

- **flat table**
 - one item per row
 - each column is attribute
 - cell holds value for item-attribute pair
 - unique key (*could be implicit*)

attributes: name, age, shirt size, fave fruit



ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

item: person

Dataset level

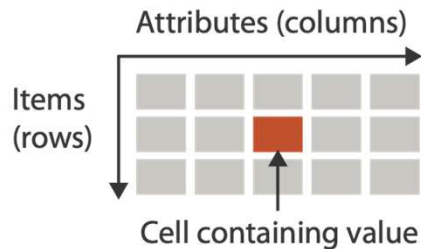
Tables

Items

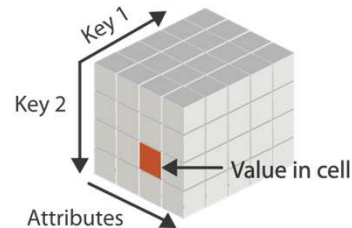
Attributes

- **multidimensional tables**
 - indexing based on multiple keys
 - eg genes, patients

→ Tables



→ Multidimensional Table



	A	B	C	D	E
1	A	B	C	D	E
2	1	A	B	C	E
3	2	A	B	C	E
4	3	A	B	C	E
5	4	A	B	C	E
6	5	A	B	C	E
7	6	A	B	C	E
8	7	A	B	C	E
9	8	A	B	C	E
10	9	A	B	C	E
11	10	A	B	C	E
12	11	A	B	C	E
13	12	A	B	C	E
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48	47	A	B	C	E
49	48	A	B	C	E
50	49	A	B	C	E
51	50	A	B	C	E
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75	74	A	B	C	E
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305	304	A	B	C	E
306	305	A	B	C	E
307	306	A	B	C	E
308	307	A	B	C	E
309	308	A	B	C	E
310	309	A	B	C	E
311	310	A			

Dataset level

- The 3D table uses three indices to find a cell

Department		A	B	C	D	E	F
Male	Accepted	512	353	120	138	53	22
	Rejected	313	207	205	279	138	351
Female	Accepted	89	17	202	131	94	24
	Rejected	19	8	391	244	299	317

Dataset availability



- There are two kinds of dataset availability: *static* and *dynamic*
- *Static file* means the entire dataset is available all at once
- *Dynamic streams* mean a dataset is continuously generated, by different sources, without having access to all the data

Attribute types: categorical vs ordered

- **Categorical (*nominal*)**

- Equal or not equal to other values
- Ex: gender, student number, course name,...
- No concept of relative relation other than inclusion in the set

- **Ordered**

- Implicit ordering
- There are two ordered types: ordinal and quantitative
 - **Ordinal** – well-defined ordering but without full-fledged arithmetic
 - **Quantitative** – well-defined capability to perform arithmetic and comparison

→ Categorical



e.g.,
fruit (apple, pear, grape),
colleges (CAMD, CCIS, COE)

→ Ordered

→ *Ordinal*



e.g.,
sizes (xs, s, m, l, xl),
months (J, F, M)

→ *Quantitative (continuous)*



e.g.,
lengths (1', 2.5', 5'),
population

Ordered attribute type: **direction**

- An ordered dataset can be *sequential* or *diverging*.
 - A *sequential* dataset has a homogeneous range from a minimum to a maximum.
 - A *diverging* dataset has data measured from a based point and extends to both ends
 - A *cyclic* dataset has its values wrapped around back to a starting point

→ Sequential



→ Diverging



→ Cyclic



Attributes: hierarchical

- An attribute may have a hierarchical structure.
- Ex:
 - Date sold products: 15/10/2022 (*weeks, months, years*)
 - A geographic attribute of a postal code may be aggregated to the level of cities, states,...

Cần Thơ		Ninh Kiều		An Hòa	
9	0	1	0	5	0
Province/ City		district/ town		ward	

Attribute types: categorical vs ordered



- Continuous
 - Data can take any value within the range
 - Ex: Number grade
- Discrete
 - Data can take only certain values
 - Ex: Letter grade (A+)
- We can convert between these using *clustering* or *thresholds*

Data semantics

- **Definition**

Semantics = Real-world meaning attached to data (Metadata).

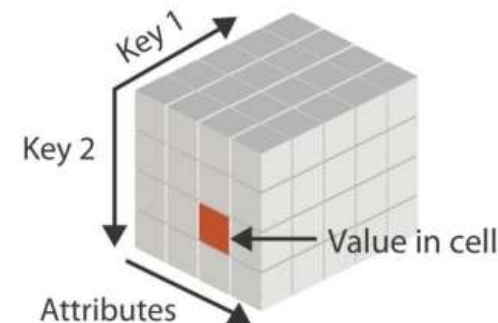
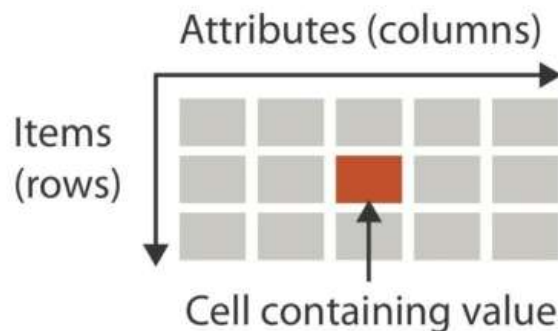
- **The "Basil 7 S Pear" Puzzle:**

- *Data:* string "Basil 7 S Pear".
- *Hypothesis:*
 - Sales: A food shipment (Basil + Pears) arriving on the 7th?
 - Services: "Basil Point" neighborhood has 7 inches of snow, cleared by "Pear Creek" company?
 - Experiment: A lab rat named "Basil" took 7 tries to run the "South" maze for a "Pear" reward?

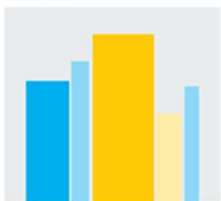
⇒ Visualization requires understanding semantics to choose the right display.

Data semantics – Key and Value

- **Key** semantics
 - A **key** attribute is an index that is used to find **value** attributes
 - In a table, the key may just be *explicit* or *implicit* (indices)
 - Keys may be *categorical* or *ordinal* attributes
- **Value** attribute is a value of the table cell

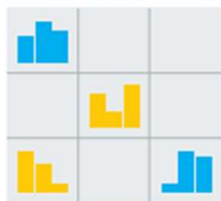


VARIABLE WIDTH
CHART



Two variables
per item

TABLE WITH
EMBEDDED CHARTS



Many
categories

BAR CHART
HORIZONTAL



One variable
per item

BAR CHART
VERTICAL



Few
categories

CIRCULAR AREA
CHART



Cyclical
data

LINE CHART



Non-cyclical
data

BAR CHART
VERTICAL



Single or few
categories

LINE CHART



Many
categories

SCATTER PLOT



Two
variables

SCATTER PLOT
BUBBLE SIZE



Three or more
variables

Among
items

Over time

COMPARISON

What would you
like to show?

RELATIONSHIP

DISTRIBUTION

COMPOSITION

Few data
points

BAR HISTOGRAM



LINE HISTOGRAM



Many data
points

SCATTER PLOT



Single
variable

Two
variables

Changing
over time

Static

Few
periods

Many
periods

Only relative
differences
matter

Relative
and absolute
differences
matter

Only relative
differences
matter

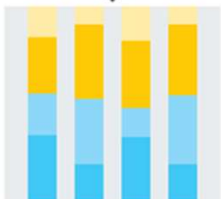
Relative
and absolute
differences
matter

Simple
share
of total

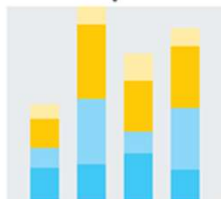
Accumulation
or subtraction
to total

Components
of components

Accumulation
to total and
absolute differ-
ence matters



STACKED 100%
BAR CHART



STACKED BAR
CHART



STACKED AREA
100% CHART



STACKED AREA
CHART



PIE CHART



WATERFALL CHART



STACKED 100%
BAR CHART WITH
SUBCOMPONENTS



TREEMAP

How to Design Vis. Idioms?

Idiom	Chart name
What	Data abstraction
How	Encoding
	Manipulate
	Facet
	Reduce
Why	Task abstraction

Mark & Channel



Marks as **items/nodes**

- A *mark* is a basic graphical element in an image.
- Marks are geometric primitives classified according to the number of dimensions they require
- Points – 0D Lines – 1D Areas – 2D Volume – 3D



Points – 0D



Lines – 1D



Areas – 2D

Channels

- A visual *channel* is a way to control the appearance of the mark, independent of the dimensionality of the geometric primitives
- Commonly seen channel types: *position*, *color*, *shape*, *tilt*, and *size*
- Other possible channels: *depth*, *luminance*, *saturation*, *curvature*, ...

Channels

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Shape



→ Tilt



also known as angle

→ Size

→ Length



→ Area



→ Volume



Visual encoding

mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin	name
18	8	307	130	3504	12	1970	USA	chevrolet chevelle malibu
15	8	350	165	3693	11.5	1970	USA	buick skylark 320
18	8	318	150	3436	11	1970	USA	plymouth satellite
16	8	304	150	3433	12	1970	USA	amc rebel sst
17	8	302	140	3449	10.5	1970	USA	ford torino
15	8	429	198	4341	10	1970	USA	ford galaxie 500
14	8	454	195	4354	9	1970	USA	chevrolet impala
14	8	440	230	4312	8.5	1970	USA	plymouth fury iii
14	8	455	225	4425	10	1970	USA	pontiac catalina
15	8	390	190	3850	8.5	1970	USA	amc ambassador dpl
15	8	360	170	3563	10	1970	USA	dodge challenger se
14	8	360	160	3609	8	1970	USA	plymouth 'cuda 340
15	8	400	150	3761	9.5	1970	USA	chevrolet monte carlo
14	8	400	225	3086	10	1970	USA	buick estate wagon (sw)
24	4	113	95	2372	15	1970	Japan	toyota corona mark ii
22	6	161	95	2833	15.5	1970	USA	plymouth duster
18	6	161	97	2774	15.5	1970	USA	amc hornet
21	6	181	85	2587	16	1970	USA	ford maverick

Item → Mark

Attribute → Channel

Visualization building blocks

→ Points



→ Lines



→ Areas



Mark



→ Position

→ Horizontal



→ Vertical



→ Both



→ Color **Channel**



→ Shape



→ Tilt



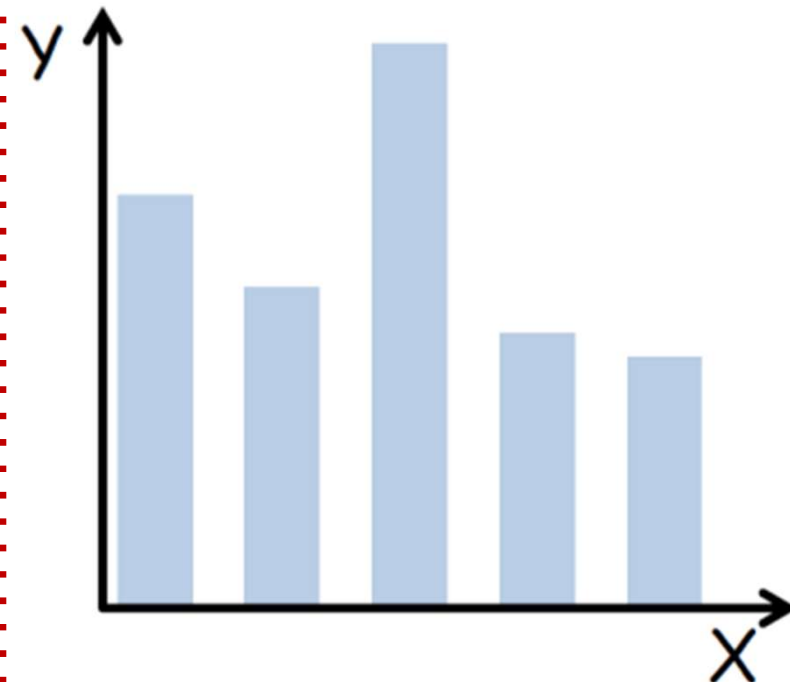
→ Size

→ Length

→ Area

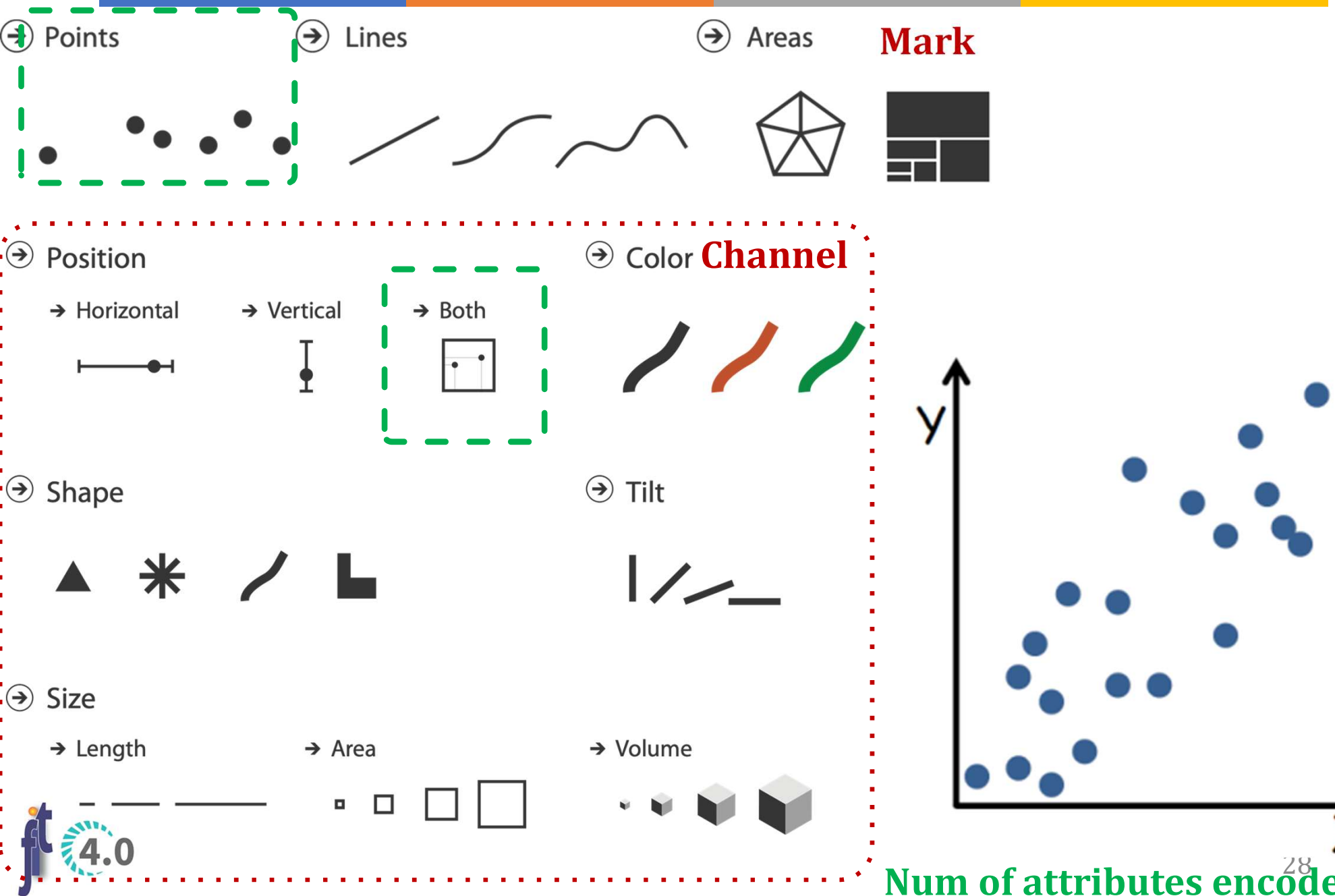


→ Volume

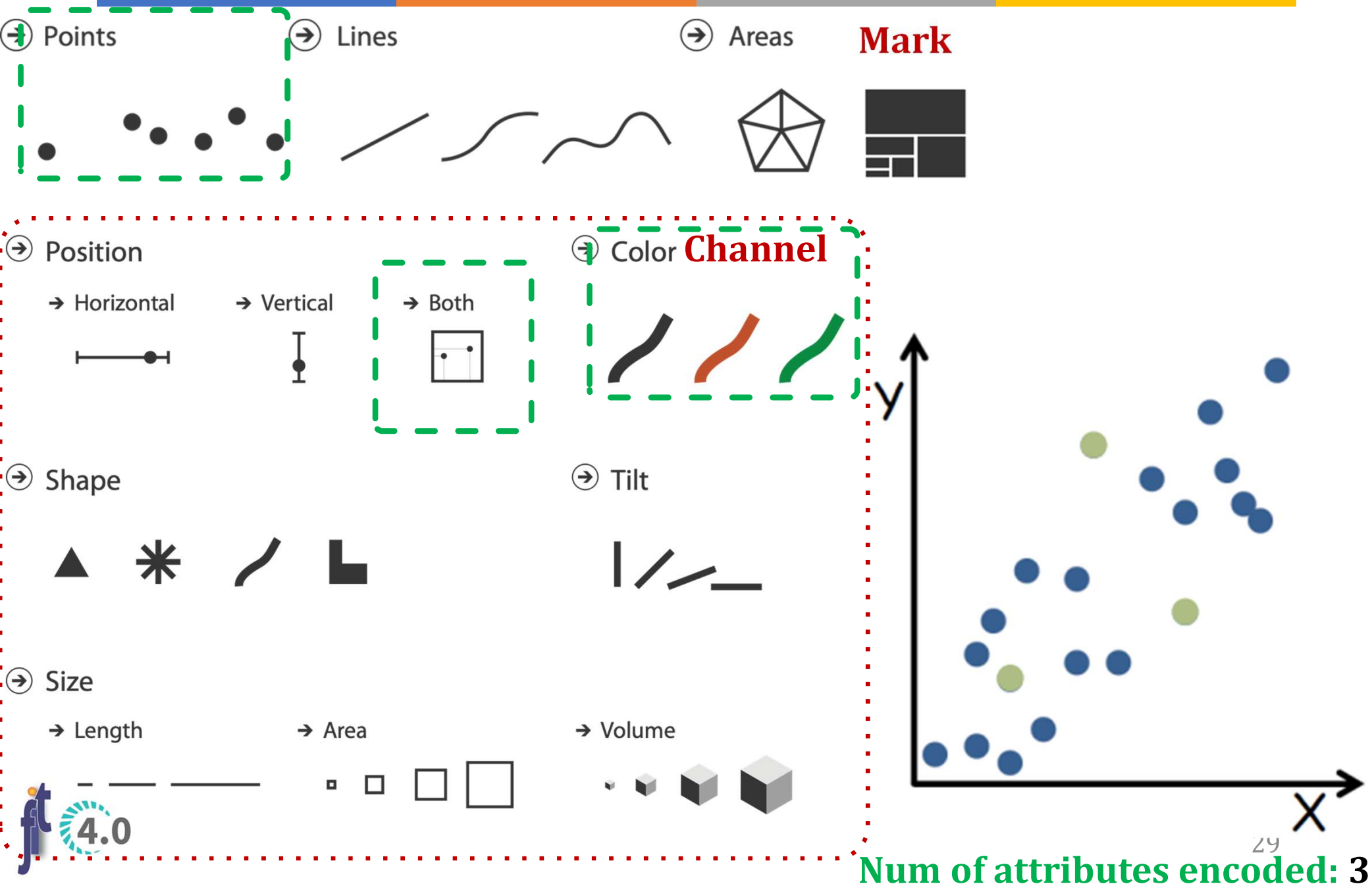


Num of attributes encoded: 2

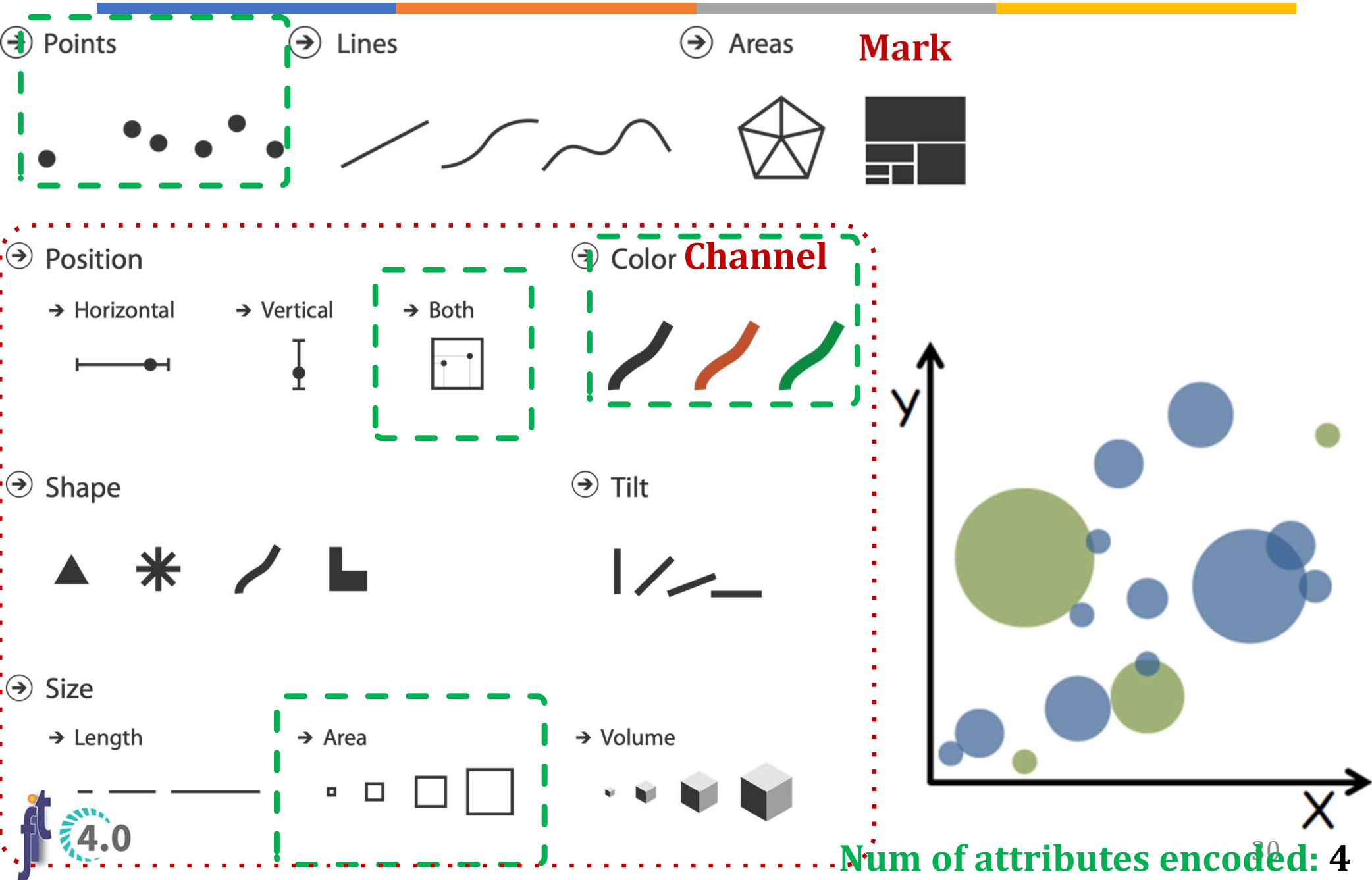
Visualization building blocks



Visualization building blocks



Visualization building blocks



Visualization building blocks

➔ Points



➔ Lines



➔ Areas



Mark



➔ Position

➔ Horizontal



➔ Vertical



➔ Both



➔ Color **Channel**



➔ Shape



➔ Tilt



➔ Size

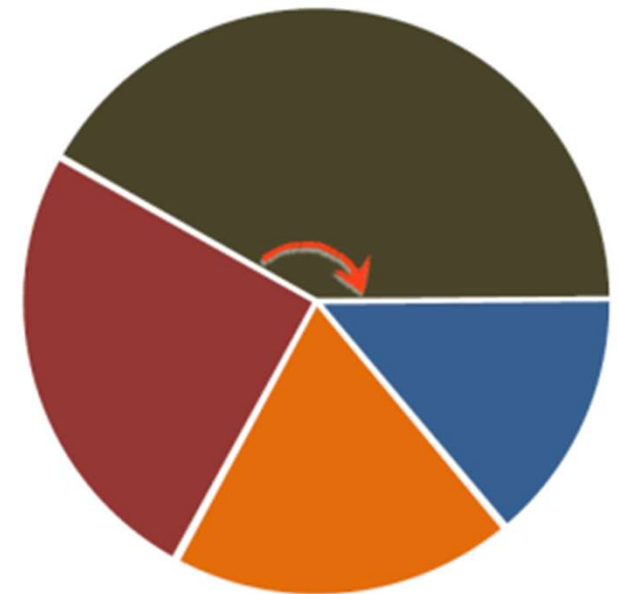
➔ Length



➔ Area



➔ Volume



Num of attributes encoded: 2

Visualization building blocks

➔ Points



➔ Lines



➔ Areas



Mark



➔ Position

➔ Horizontal



➔ Vertical



➔ Both



➔ Color **Channel**



➔ Shape



➔ Tilt



➔ Size

➔ Length



➔ Area



➔ Volume



Num of attributes encoded: 2

Visualization building blocks

➔ Points



➔ Lines



➔ Areas



Mark



➔ Position

➔ Horizontal



➔ Vertical



➔ Both



➔ Color **Channel**



➔ Shape



➔ Tilt



➔ Size

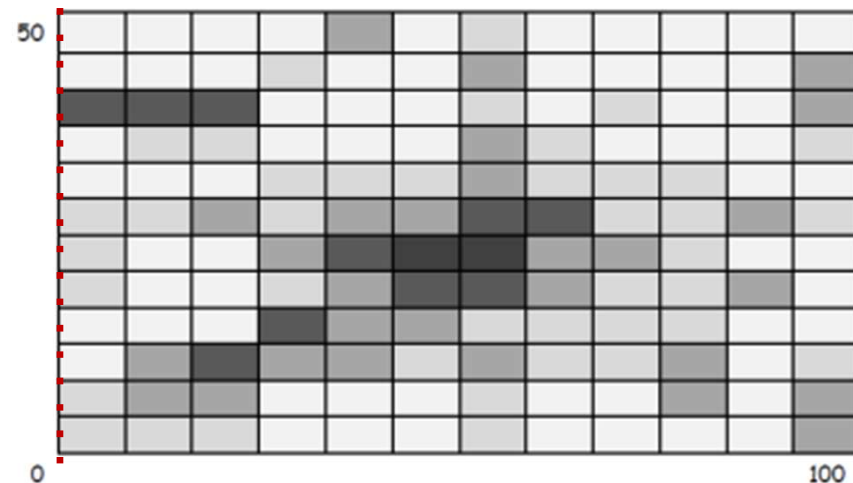
➔ Length



➔ Area



➔ Volume



Num of attributes encoded: 3

Color



Big picture

Encode > Map

② Color

→ Color Encoding



→ Color Map

→ Categorical



→ Ordered

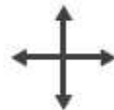
→ Sequential



→ Diverging



→ Bivariate



② Size, Angle, Curvature, ...

→ Length



→ Angle



→ Area



→ Curvature



→ Volume



② Shape



② Motion

→ Motion Direction, Rate, Frequency, ...



Color spaces

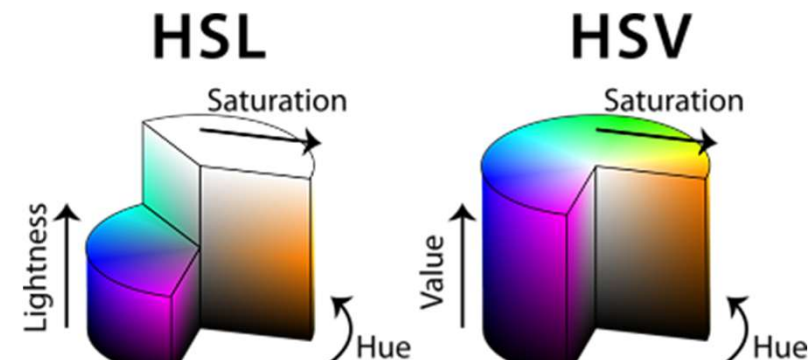
- The **color space** of what colors the human visual system can detect is three-dimensional
- Some of these color spaces are for *computer manipulation*, and the others are a better match with the characteristic of *human vision*
- The most common color space is the **RGB** system
 - Each color is represented by the weights assigned to the **red**, **green** and **blue** channels.

Color spaces

- **HSL system**

- Requires HUE, Saturation, and Lightness axis
- Each color is defined by the weights assigned to the HUE, saturation and lightness axis
 - *HUE* axis captures what we normally think of as *pure color*
 - Not mixed with pure white and pure black
 - *Saturation* axis is the amount of *white mixed* with a pure color
 - Pink = red + pure white
 - *Lightness* axis is the amount of *black mixed* with a color

- **HSV space** is similar, where V is for *grayscale* and is linear to lightness L



Luminance, Saturation and HUE Channels

- *Luminance*, *saturation* and *HUE* can be used as a *magnitude* channel and as an *identity* channel
 - Magnitude channel
 - Luminance
 - Saturation
 - Identify channel
 - HUE

Luminance, Saturation and HUE Channels

- *Magnitude channel*

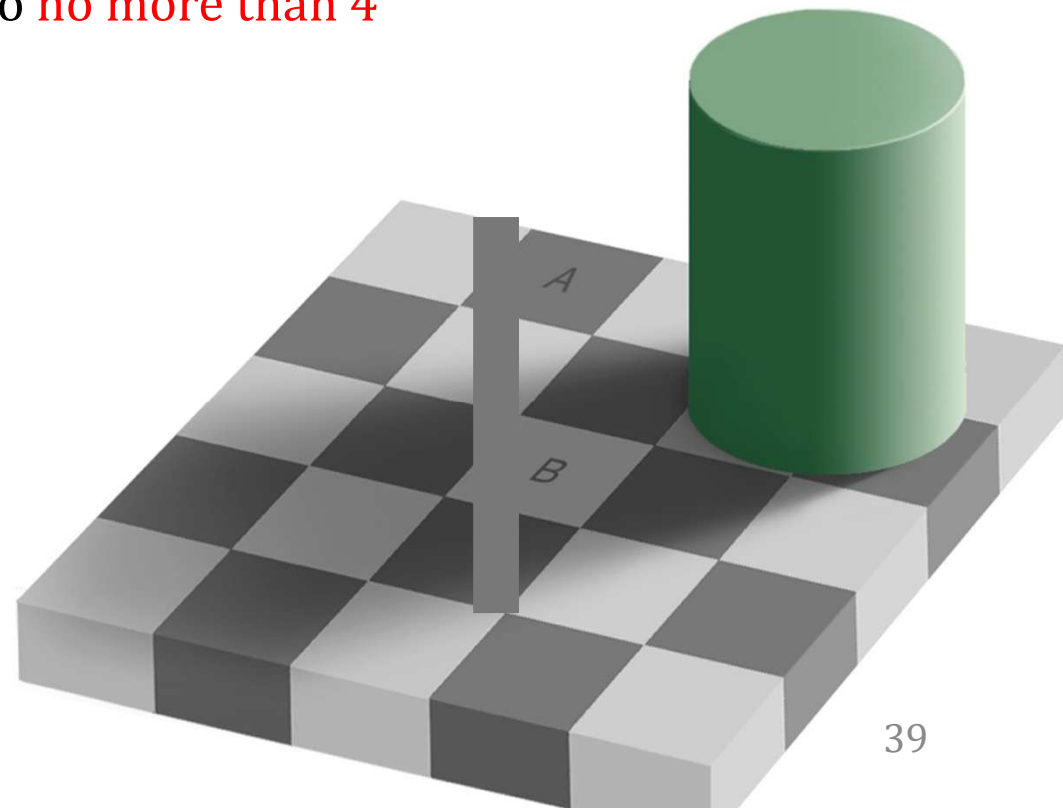
- *Luminance*

- Suitable for **ordered data**
 - Low-accuracy channel in terms of perceiving whether noncontiguous regions have the same luminance because of contrast effects
 - Limit grayscale bins to **no more than 4**

- Saturation

- *Identify channel*

- HUE



Luminance, Saturation and HUE Channels

- *Magnitude channel*

- *Luminance*

- *Saturation*

- Suitable for **ordered data**
 - Low-accuracy for noncontiguous regions
 - The number of discriminate steps for saturation is low: **around 3 bins**
 - Interact with the size channel
 - Use bright, **highly saturated colors so that small regions** are more distinguishable

- *Identify channel*

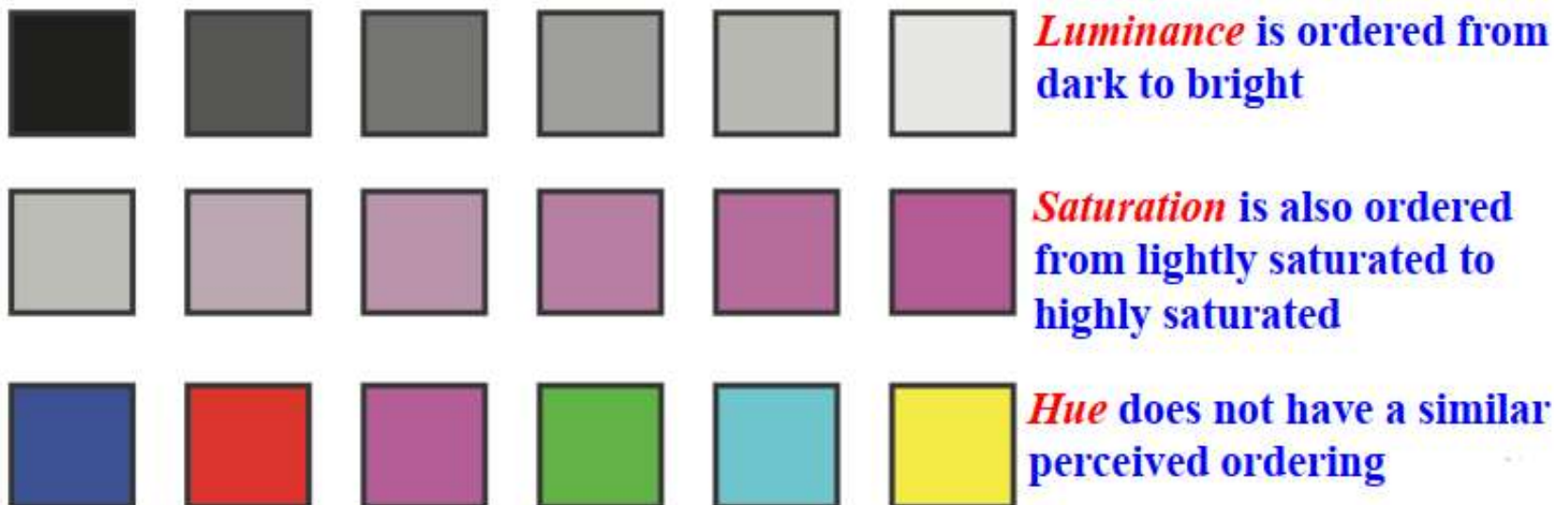
- *HUE*

Luminance, Saturation and HUE Channels

▪ *Identify channel*

○ *HUE*

- Very effective for **categorical data** and showing groups (*highest ranked after **spatial position***)
- HUE and size can interact: HUE is harder to distinguish in small regions
- Make fine distinctions in HUE for contiguous regions
- Very limited discriminability between separate regions
- HUE does not have an implicit perceptual ordering



Colormaps



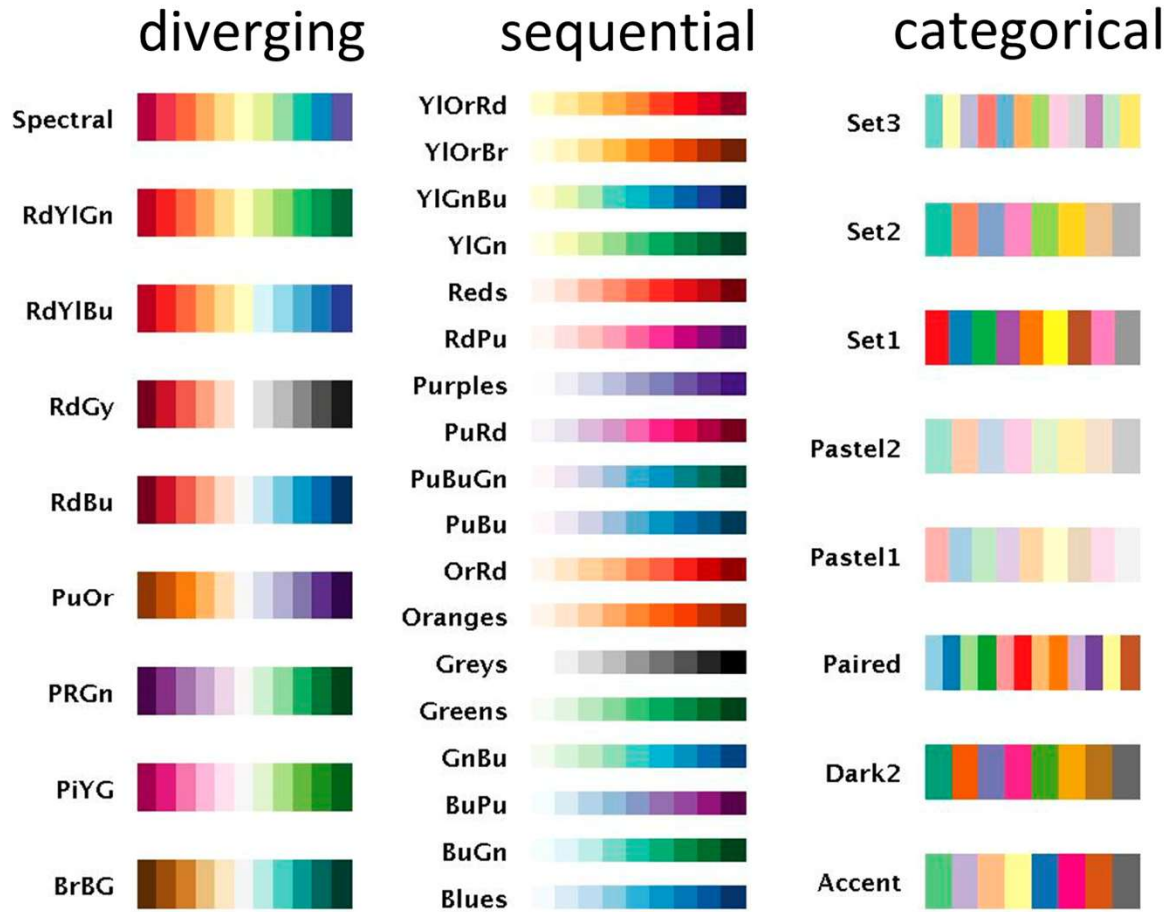
Colormaps

- A **colormap** defines a mapping between colors and data values (*visual encoding with colors*)
- **Colormaps** can be *categorical* or *ordered*
 - Ordered colormaps can be *sequential* or *diverging*
- **Colormaps** can also be a *continuous* range of values or *segmented* into discrete bins of colors
 - *Continuous colormap*s are used for showing *quantitative* attributes
 - *Segmented colormap*s are suitable for *categorical* data
 - For *ordinal* data
 - *Segmented colormaps* will emphasize their *discrete* nature
 - While *continuous colormaps* will emphasize their *ordered* nature

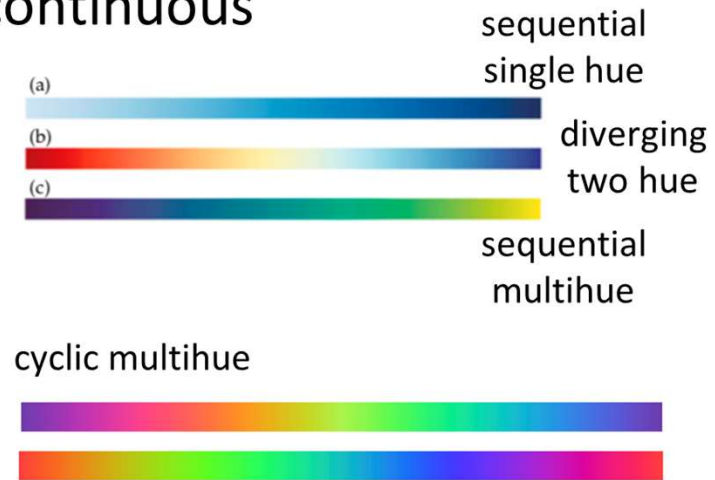
Colormaps



segmented



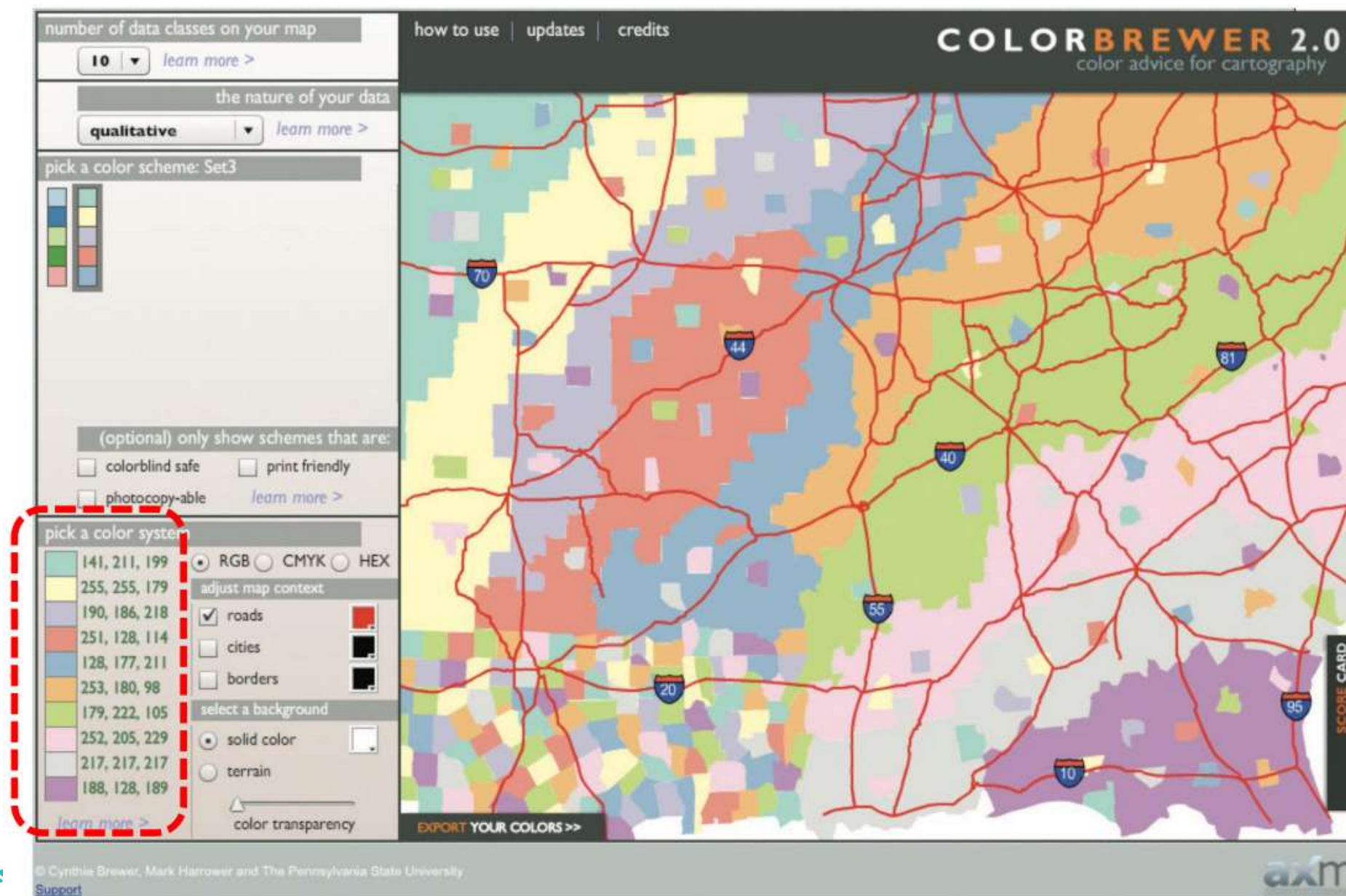
continuous



Categorical Colormaps

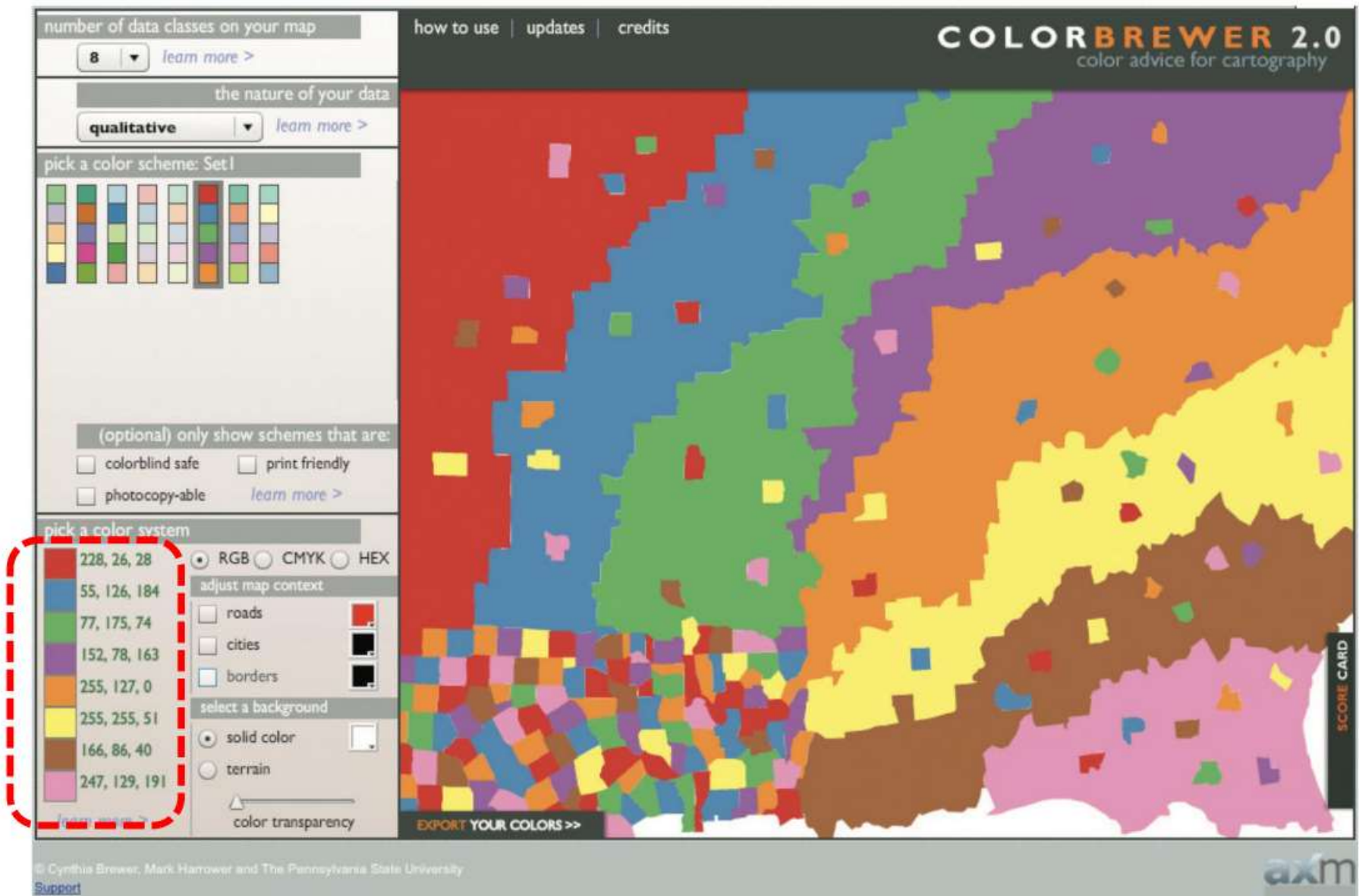
- For other uses, colors should be sufficiently different in luminance that they can be distinguished even in black and white
- ***Colormaps*** for
 - *Small regions* such as lines should be *highly saturated*
 - *Large regions* such as areas should have *low saturation*
- An appropriate ***colormap*** may depend on the *mark type* being used.

Categorical Colormaps - Example



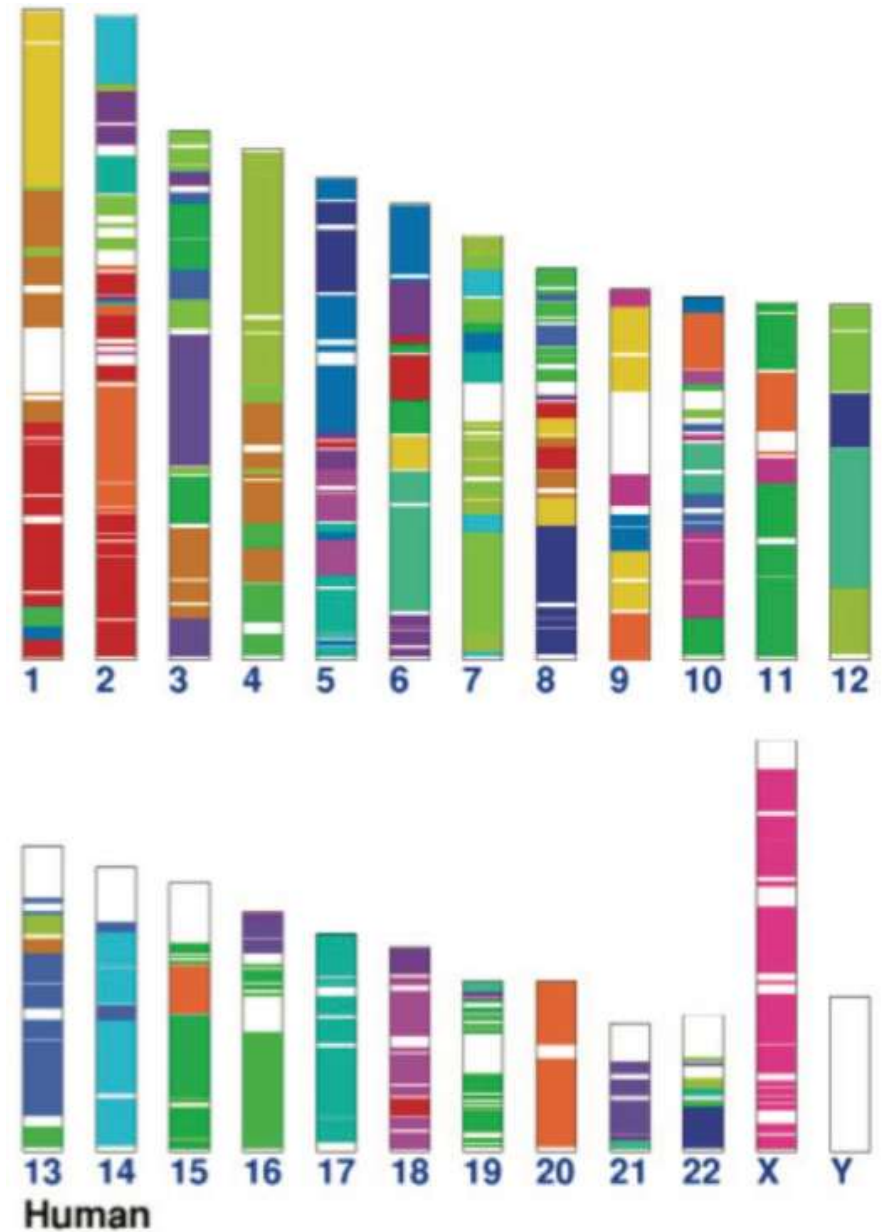
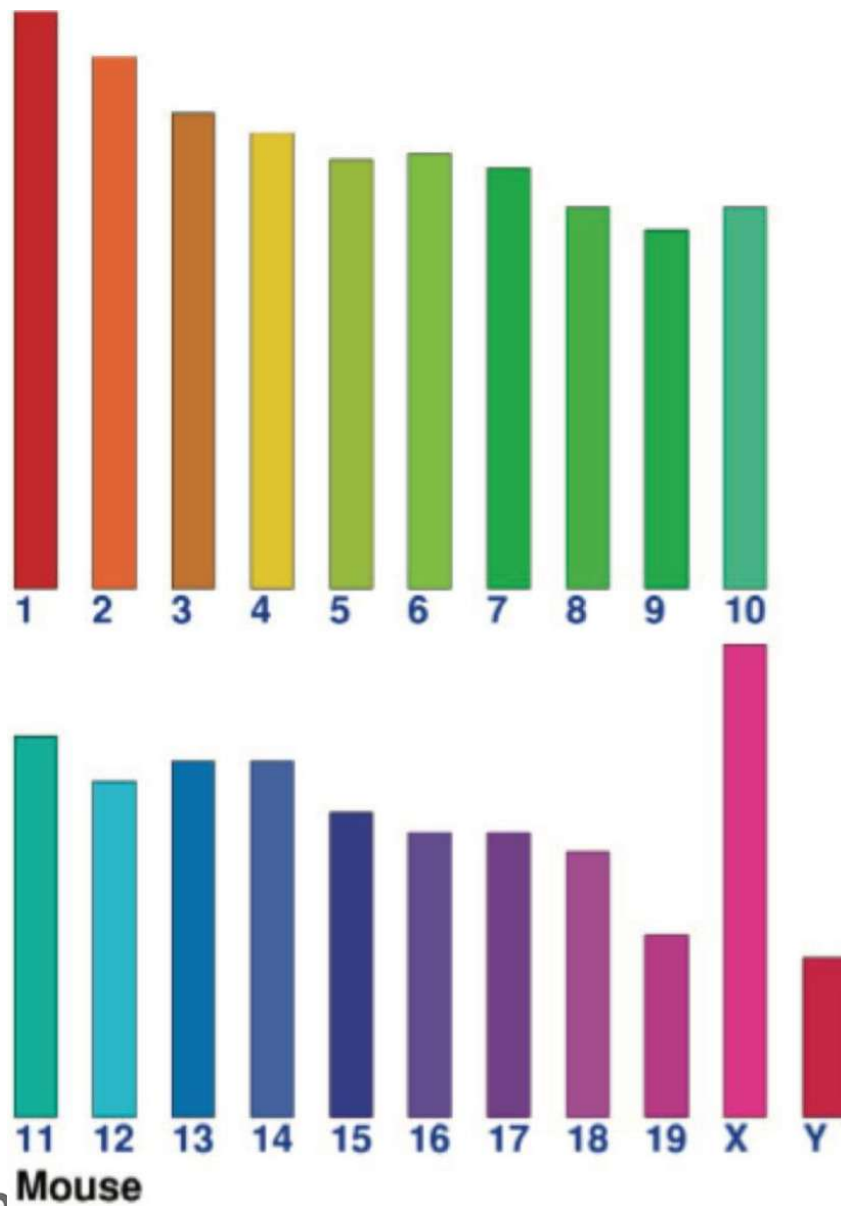
This 10-color low saturation map works well with large areas.

Categorical Colormaps - Example



**This 8-color high saturation map works poorly with large areas.
It would be better suited for small regions.**

Categorical Colormaps - Example



Ordered Colormaps

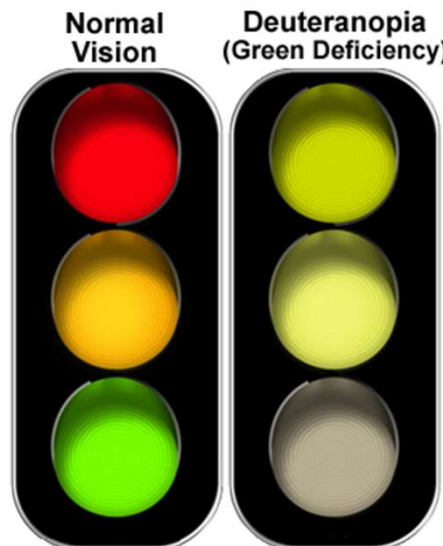
- An *ordered* colormap is appropriate for encoding *ordinal* or *quantitative* attributes
- The two major variants of *continuous colormaps* for ordered data have expressiveness characteristics that should match up with the attribute type.
 - Sequential
 - Diverging

Ordered Colormaps

- A *sequential* colormap
 - Ranges from a minimum value to a maximum value
 - If only the *luminance* channel is used, the result is a grayscale ramp
 - When *hue* is added, one end of the map is a specific hue at full saturation and brightness
 - If *saturation* is a variable, the other end is pale or white
 - If *luminance* is the varying quantity, the other end is dark or black
- A *diverging* colormap
 - has two hues at the endpoints and a neutral color as a midpoint
 - This neutral color may be white, gray, or black, or a high-luminance color such as yellow.

Colorblind-Safe Colormaps

- **Color blindness** is a sex-linked inherited trait that affects 8% of males and 0.5% of females.
 - In the common forms of color blindness, the ability to sense along the *red-green* color axis is limited or absent.
 - Many pairs that are discriminable to people with normal color vision are confused
 - Including *red-black*, *blue-purple*, *light green-white*, and *brown-green*.



Colorblind-Safe Colormaps

- Avoid the use of only the HUE channel to encode information
 - In addition to HUE design, categorical colormaps vary in luminance or saturation
- If possible, avoid colormaps that emphasize red-green
 - Especially in divergent red-green ramps

Expressiveness & Effectiveness principles

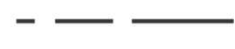


Expressiveness

➔ Magnitude Channels: **Ordered** Attributes

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance  Same

Color saturation  Same

Curvature  Same

Volume (3D size)  Same

➔ Identity Channels: **Categorical** Attributes

Spatial region 

Color hue 

Motion 

Shape 

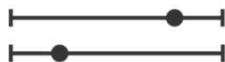
Effectiveness

- The *effectiveness* principle states that
 - The importance of the attribute
 - Match the *salience* of the channel
 - Encoded with the most effective channels in order to be most noticeable
 - Less important attributes
 - Matched with less effective channel

Effectiveness

➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)



Same

Best
Effectiveness
Least

➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion



Shape



- *Spatial channels* are the only ones that appear on both lists at the top
- *None of the others* are effective for both data types
- The above channels are for 2D, and 3D depth is a much lower channel

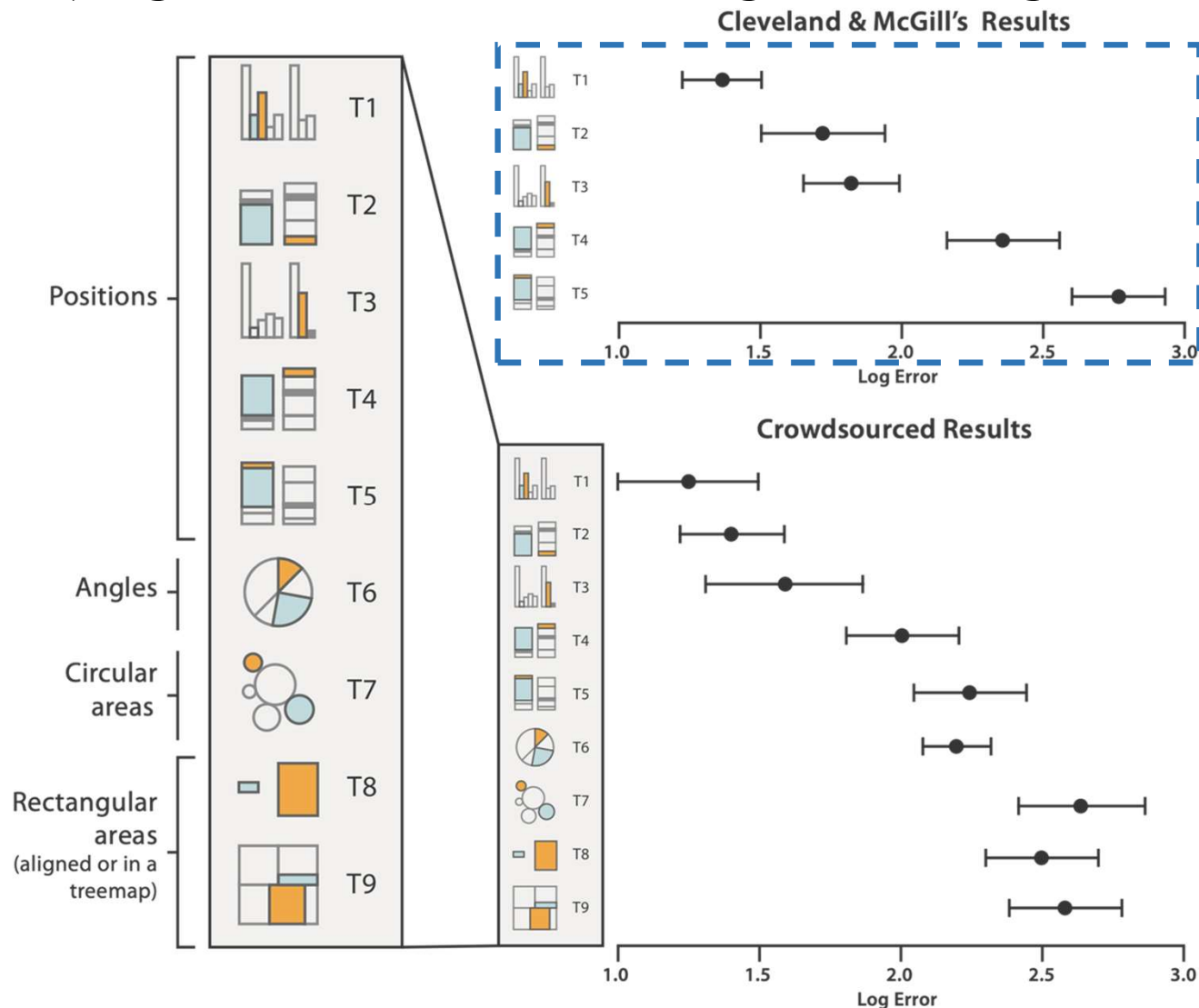
Channel effectiveness

- How do we identify the *effectiveness* of a *channel*?
 - The criteria of effectiveness channel
 - *Accuracy*
 - *How precisely can we tell the difference between encoded items?*
 - *Discriminability*
 - *How many unique steps can we perceive?*
 - *Separability*
 - *Is our ability to use this channel affected by another one?*
 - The ability to provide *visual popout*
 - *Can things jump out using this channel?*
 - The ability to provide *perceptual grouping*

Accuracy

Results largely confirmed Cleveland & McGill (1984) with respect to the relative order.

The area judgment tasks are shown to give even larger errors



$T1 < T2 < T3 < T4 < T5 < T6 < T7 < T8 < T9 < T7$

Discriminability

- **Discriminability**

- If a data attribute is encoded using a particular visual channel, are the differences between items (of this attribute) perceptible to the human as intended?
- The characterization of a visual channel should quantify the number of **bins** that are available for use within a visual channel, where each **bin** is a distinguishable step

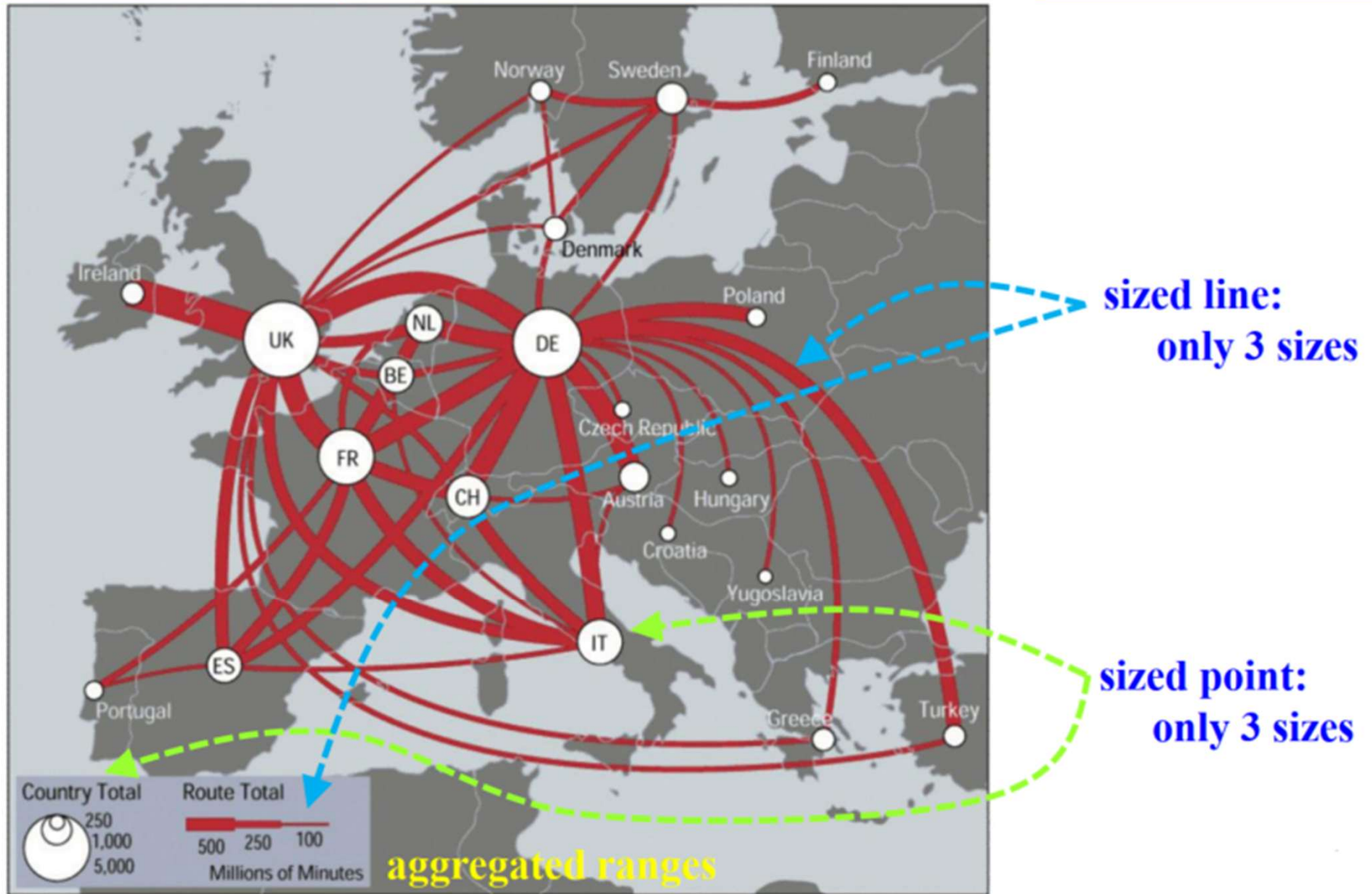
- **EX:**

- Changing *line width* past a certain limit may be perceived *as an area* rather than a line mark

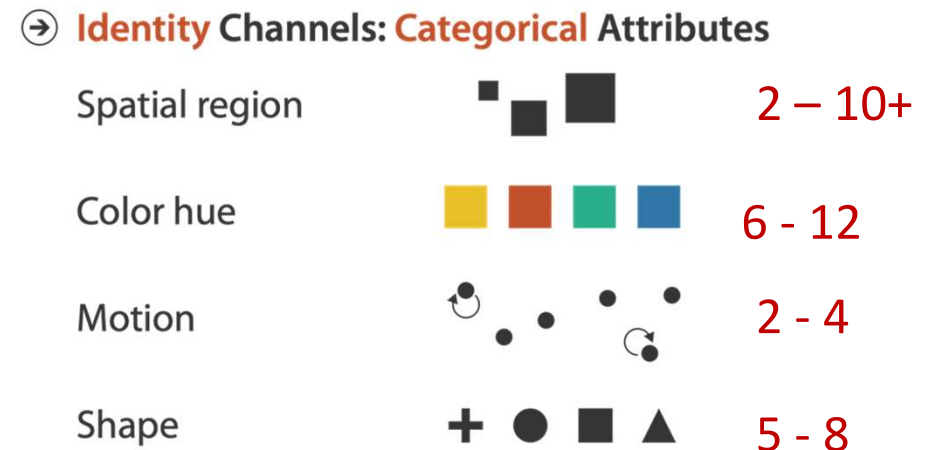
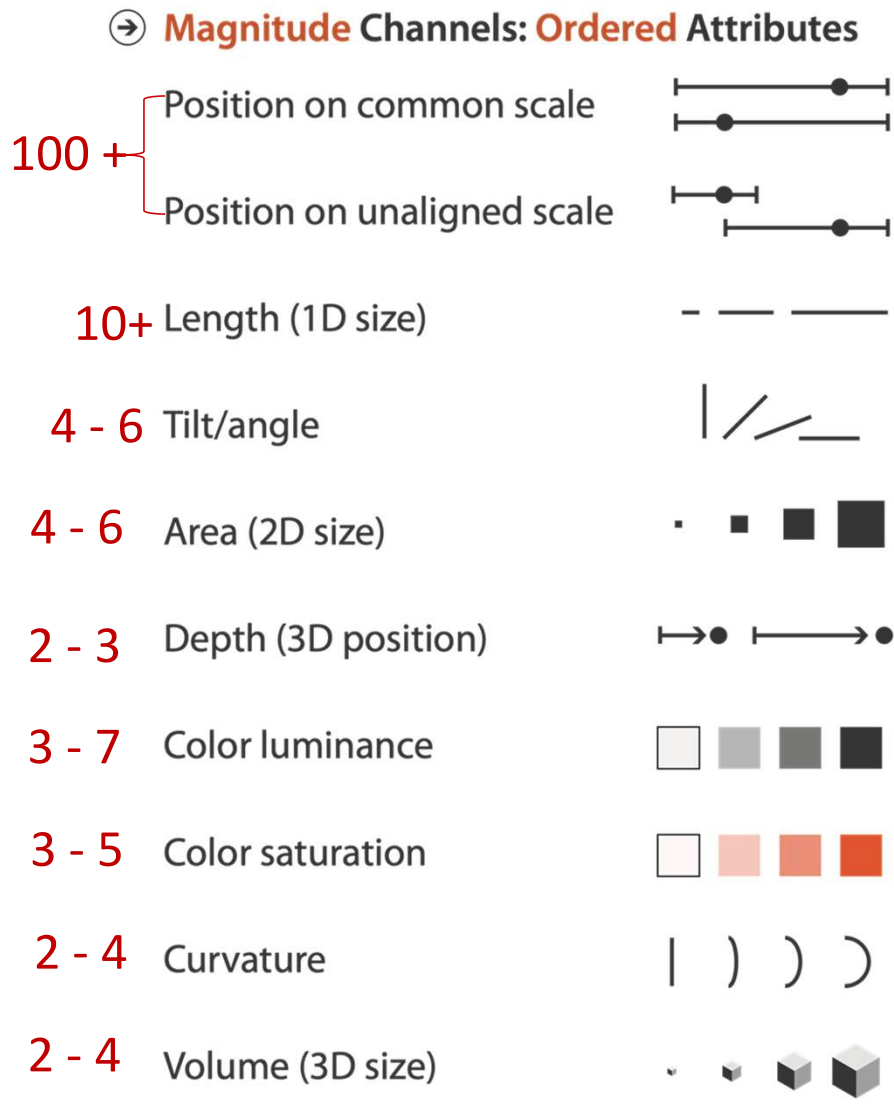
- *Line width* may be very usable for a small number of different values

- But would be a poor choice for *dozens* or *hundreds* of values

Discriminability



Discriminability

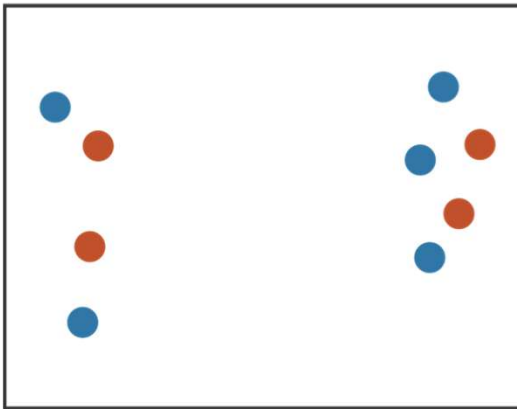


Separability

- Visual channels are not completely independent of each other
 - Because some have *dependencies* and *interactions* with others.
- Consider *potential interaction* between visual channels
 - Ranging from the orthogonal and independent *separable* channels to the inextricably combined *integral* channels.
- Visual encoding separable channels is straightforward
 - But encoding different information in integral channels will fail

Separability

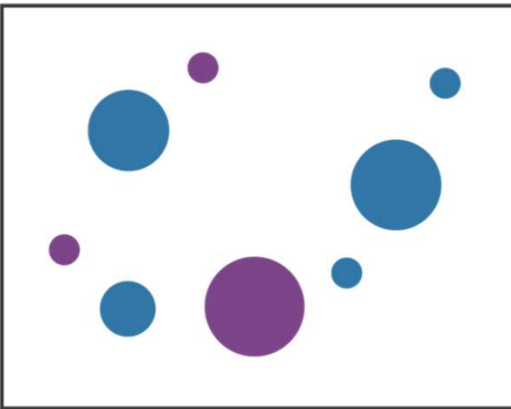
Position
+ Hue (Color)



Fully separable

2 groups each

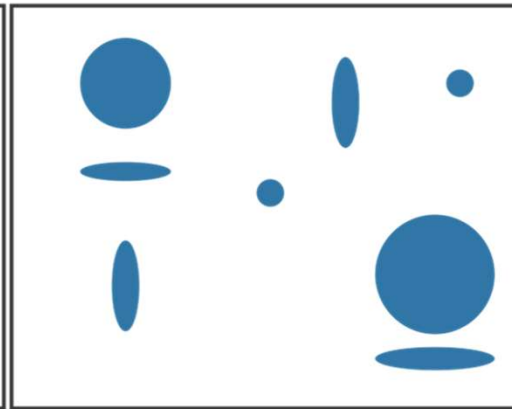
Size
+ Hue (Color)



Some interference

2 groups each

Width
+ Height



Some/significant
interference

3 groups total:
integral area

Red
+ Green



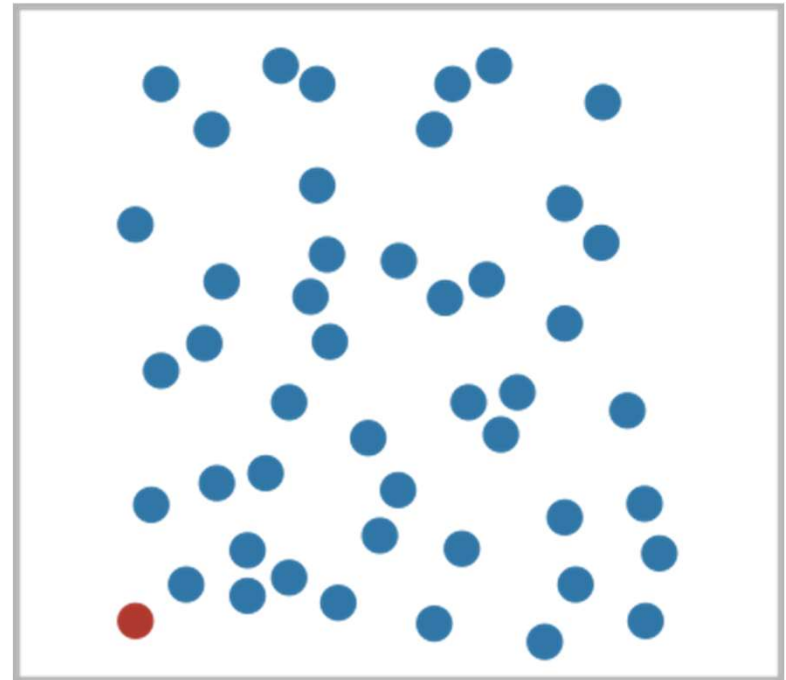
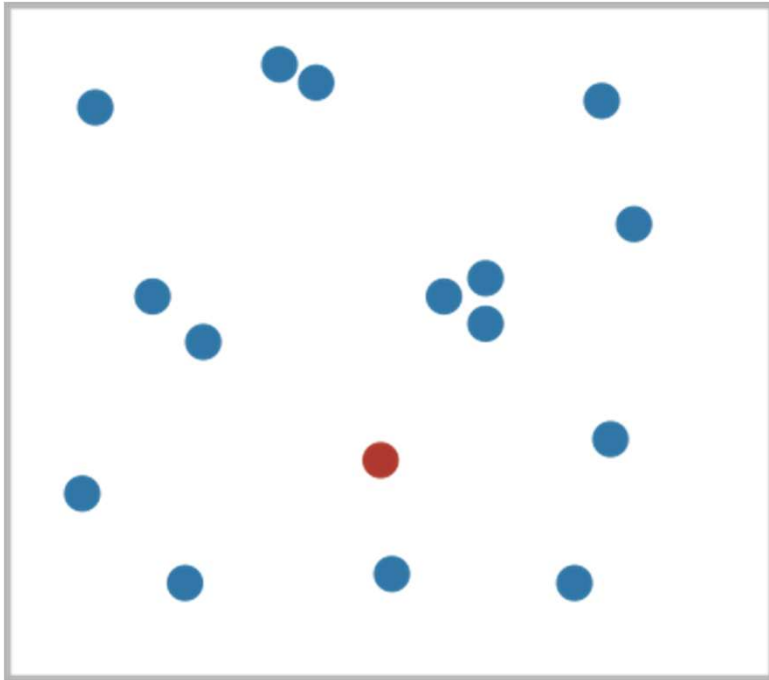
Major interference

4 groups total:
integral hue

Visual Popout

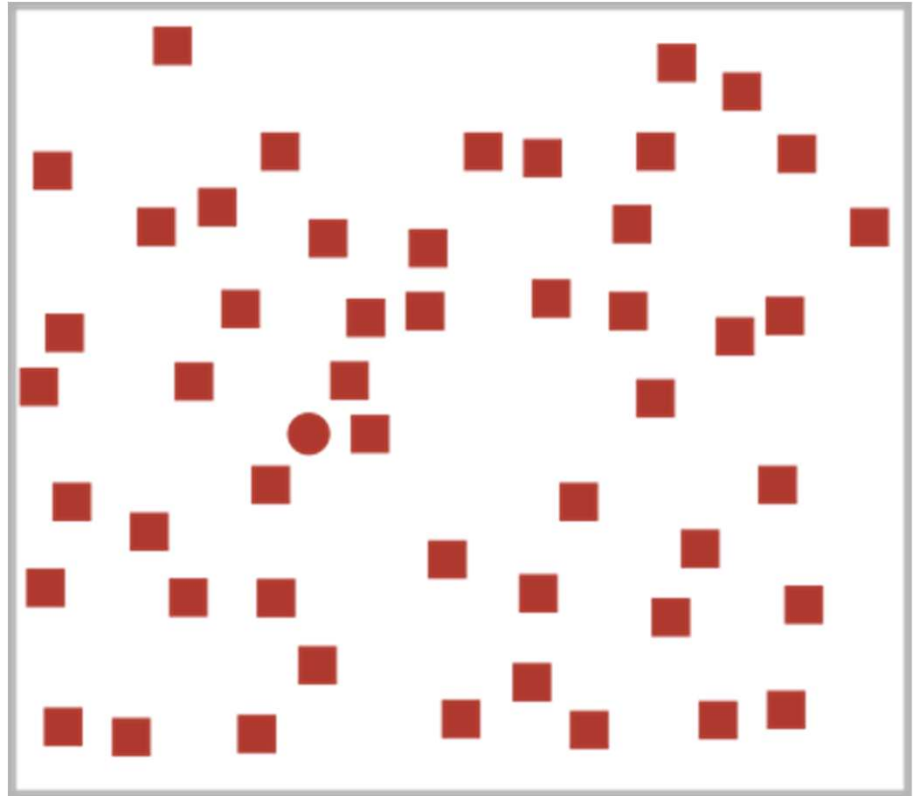
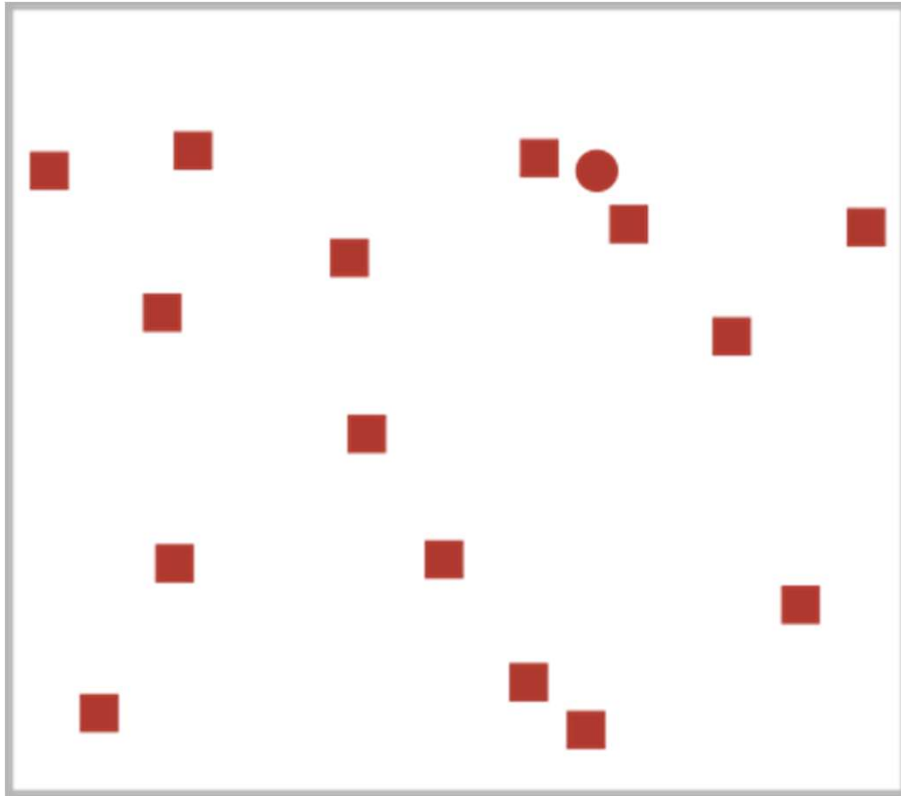
- **Visual Popout**: Making a distinct item stands out from many others immediately.
- **Popout** is also known as *pre-attentive processing* or *tunable detection*.
- Our low-level visual system can help spot in parallel the different objects from many distracting objects, without the need for the viewer to consciously direct attention to the item one by one.
- In fact, this capability is independent of the number of distracting items.

Visual Popout



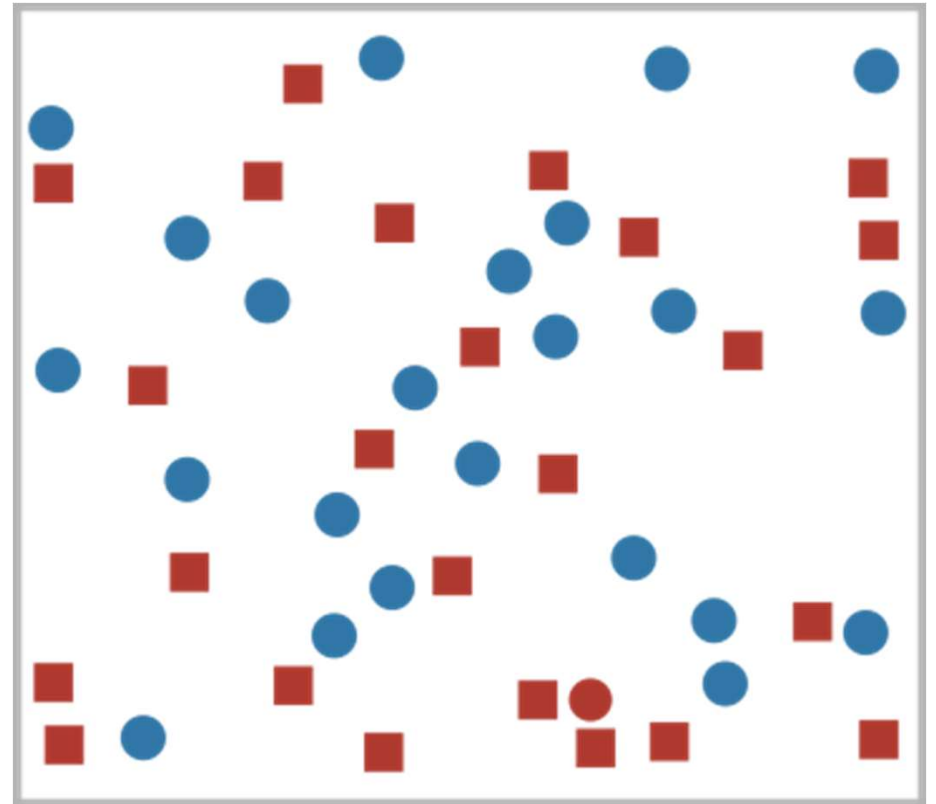
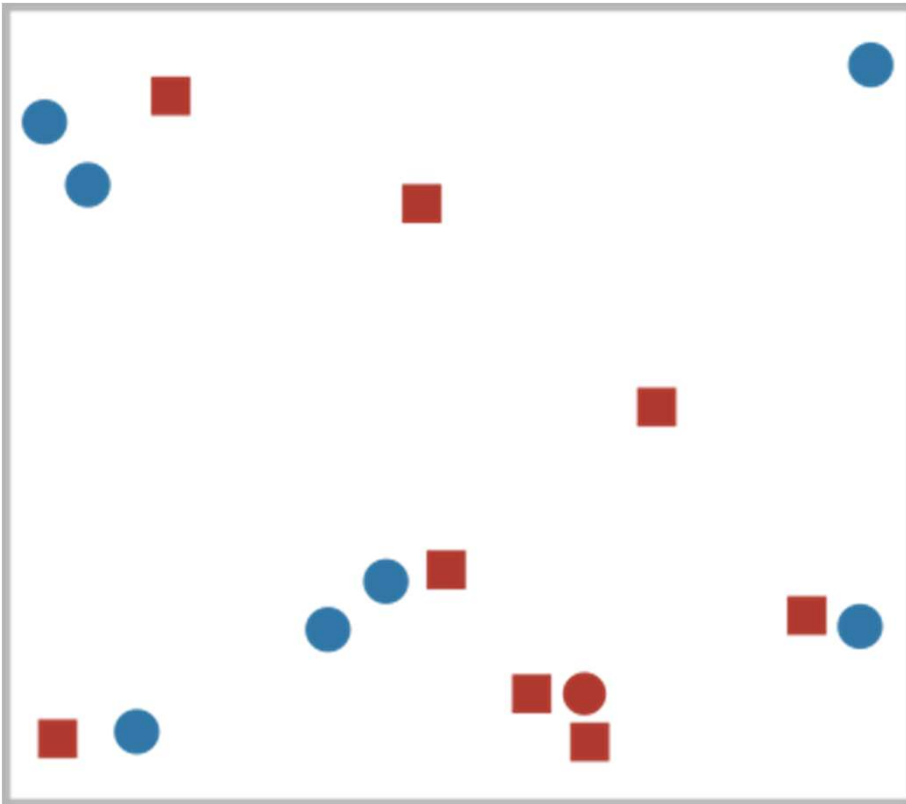
**The red circle pops out from a small set of blue circles.
We recognize this instantly**

Visual Popout



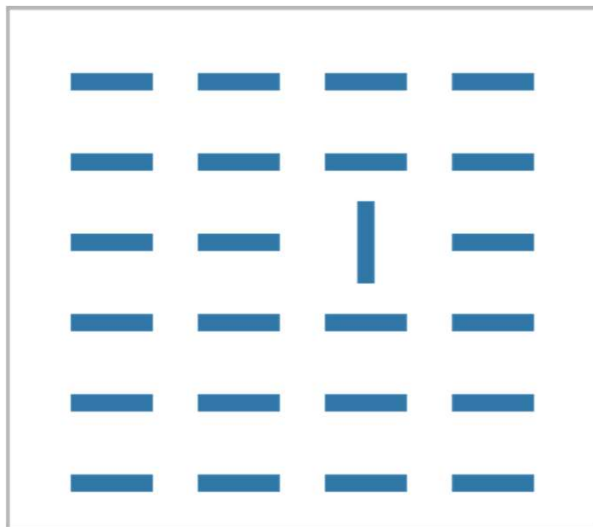
The red circle pops out from a few squares
It is a bit slower than with color, though.

Visual Popout

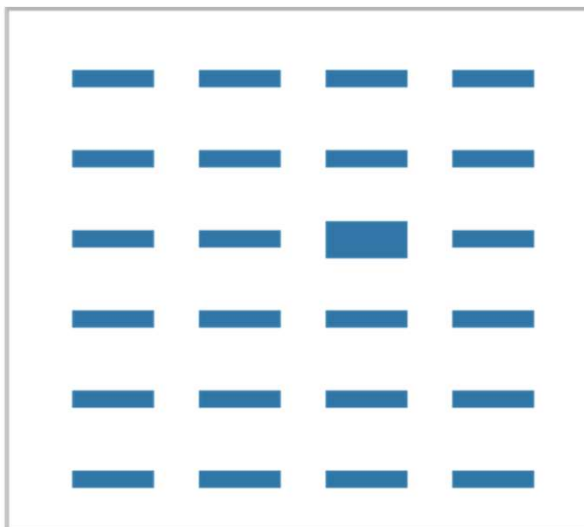


The red circle also pops out from a small set of mixed shapes and colors.

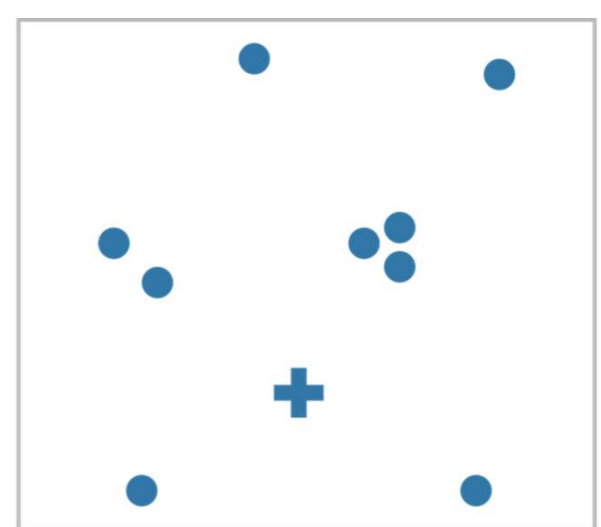
Visual Popout



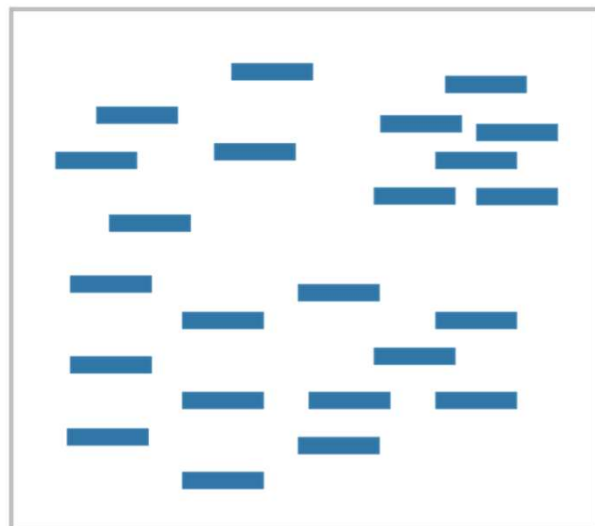
Tilt



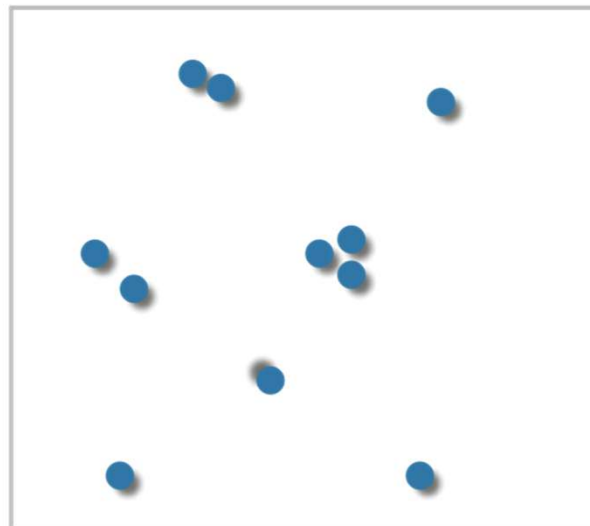
Size



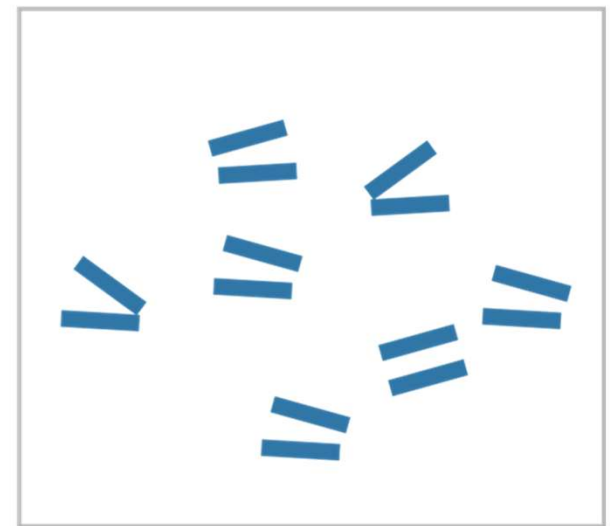
Shape



Proximity



Shadow Direction



Tilted line pairs

Many channels support visual popout

Ví dụ

- “Quản lý muốn biết **thể loại sách nào đang mang lại doanh thu cao nhất** và **mối tương quan giữa giá sách và số lượng bán ra của từng cuốn sách** để quyết định chương trình khuyến mãi tiếp theo.”
- **Bảng dữ liệu gồm:**
 - Mã_sách
 - Tên_sách
 - Thể_loại (Văn học, Khoa học, Thiếu nhi)
 - Giá_tiền (VNĐ)
 - Số_lượng_bán (cuốn)
 - Doanh_thu (VNĐ).

Data abstraction

- **Data type:** item/attribute
- **Dataset type:** table
- **Dataset availability:** static
- **Item semantic:** mỗi dòng là một cuốn sách cụ thể

Attribute Name	Type (C,O,Q)	Key/ value	Direction	Hierarchical	Continuous / Discrete	Semantic
Thể loại	C	Val			Rời rạc	Tên thể loại
Giá tiền	Q	Val	Tuần tự		Liên tục	Giá bán sp
Số lượng bán	Q	Val	Tuần tự		Rời rạc	Số lượng bán sp
Doanh thu	Q	Val	Tuần tự		Liên tục	Doanh thu

Task abstraction



- **Task 1**: *Tìm thể loại sách mang lại doanh thu cao nhất*

⇒ So sánh + giá trị lớn nhất

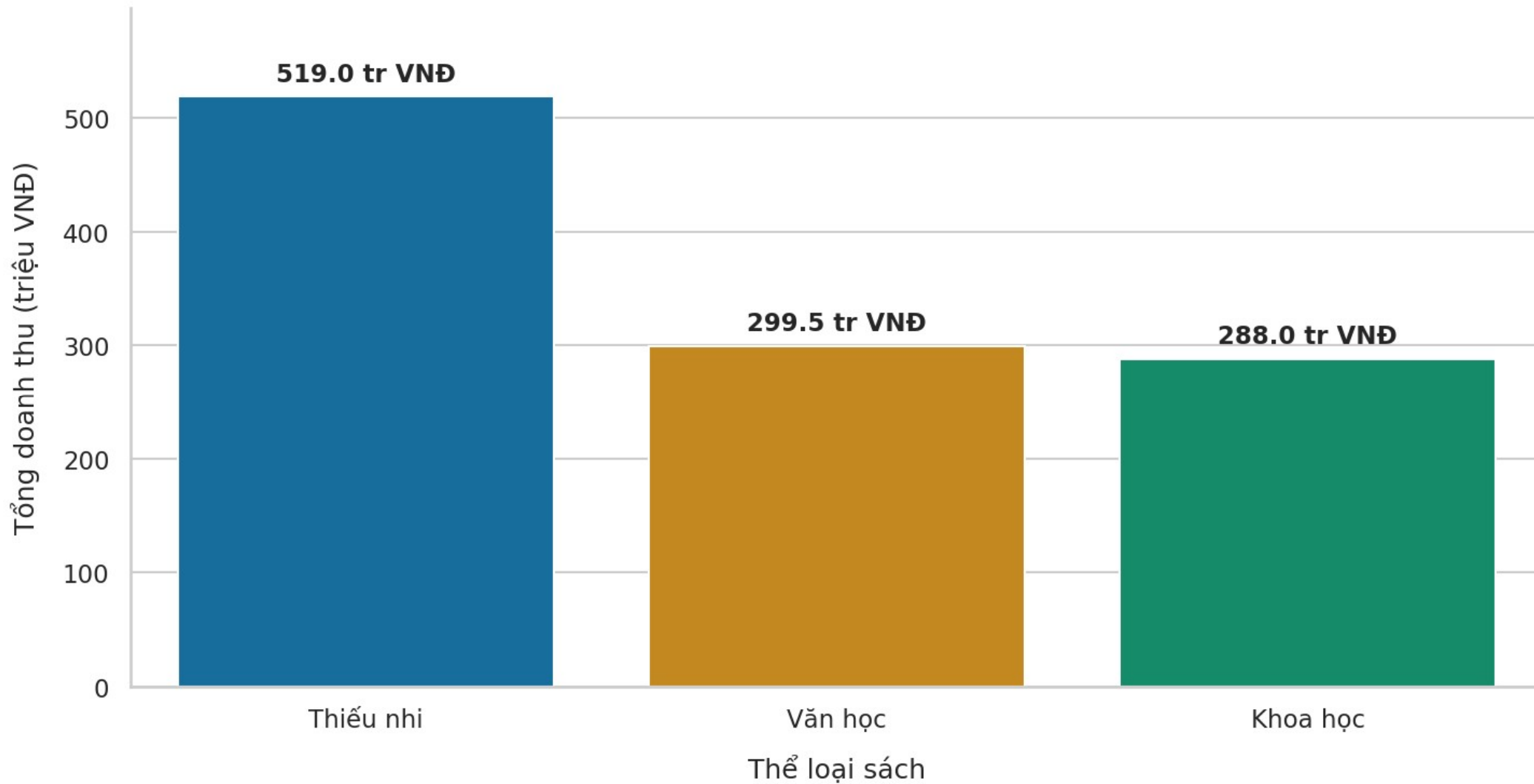
- **Task 2**: *Tìm mối tương quan giữa giá sách và số lượng bán ra của từng cuốn*

=> Tương quan + 2 thuộc tính (giá vs số lượng)

Idioms

Idiom	Bar chart
What	- Thẻ loại (1) - Tổng doanh thu (1)
How	<u>Encoding</u> Mark: line
	Channel - Pos x: <i>tổng doanh thu</i> - Region y: <i>thẻ loại</i> - Color: <i>thẻ loại</i>
Why	Compare + extreme

Vẽ biểu đồ



Đánh giá kỹ thuật

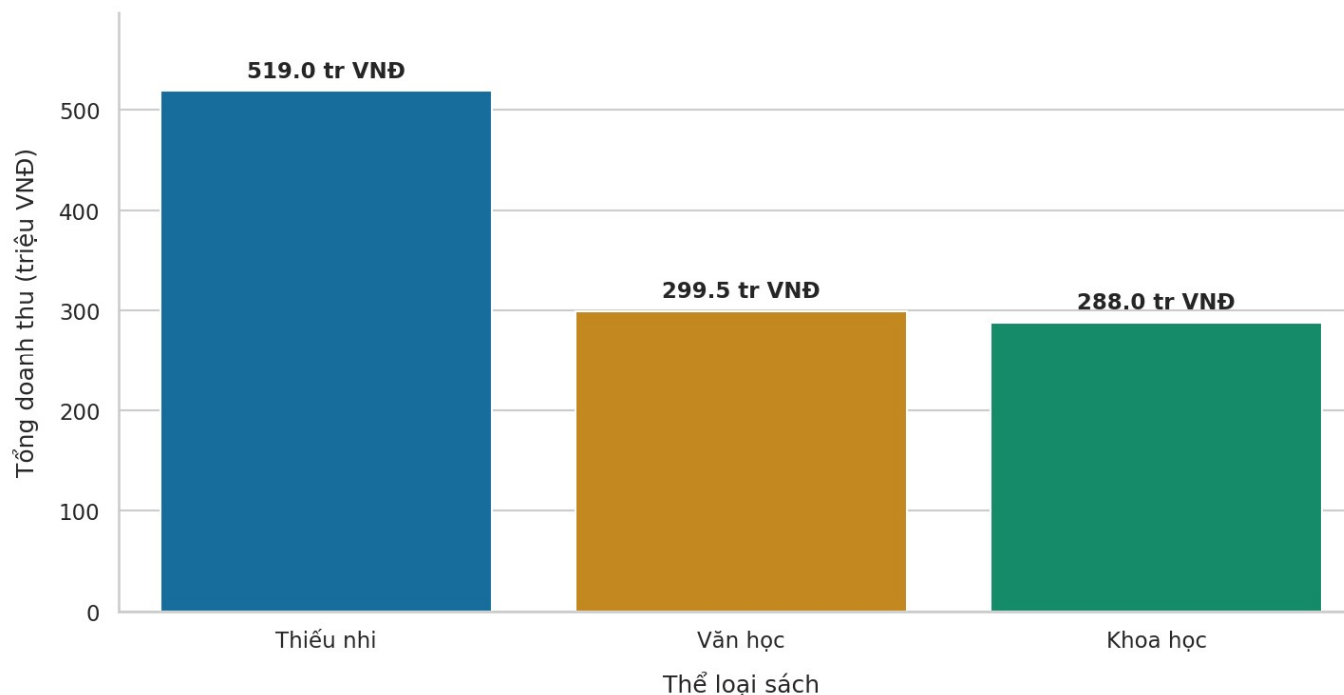
Biểu đạt; $C \rightarrow \text{region}$; $Q \rightarrow \text{position}$

Hiệu quả:

1. Độ chính xác (Accuracy): T1
2. Khả năng phân biệt (Discriminability):
 - 3 region (10+)
 - 3 color (6-12)
 - pos y (100+)
3. Khả năng phân tách (Separability):
 - Vị trí + màu sắc $\rightarrow$ phân tách
 - Vị trí dọc + vị trí ngang $\rightarrow$ phân tách
4. Khả năng nổi bật thị giác (Visual Popout)
5. Khả năng gom nhóm nhận thức (Perceptual Grouping)

Đánh giá nhận thức

Task 1: *Tìm thể loại sách mang lại doanh thu cao nhất*



Thể loại **thiếu nhi** dẫn đầu doanh thu vượt trội với **519.0 triệu VNĐ** (chiếm khoảng 47% tổng doanh thu), vượt xa hai thể loại còn lại.

Thiết kế dashboard và đánh giá



1. Phân tích nhiệt độ theo không gian & địa hình

- So sánh *nhiệt độ trung bình* giữa các vùng trên toàn quốc
- So sánh nhiệt độ trung bình/cao nhất/thấp nhất giữa các loại địa hình.
- So sánh tỉnh ven biển & nội địa.
- Thống kê mật độ điểm đo phân bố theo từng khu vực.

2. Phân tích biến động theo thời gian & thời tiết

- Xu hướng biến động nhiệt độ theo mùa
- Tần suất xuất hiện của các trạng thái thời tiết và nhiệt độ theo trạng thái thời tiết.
- So sánh thời tiết giữa các vùng.

3. Phân tích bức xạ mặt trời & thiên văn

- Tương quan giữa chỉ số UV và nhiệt độ.
- Tương quan giữa độ dài ban ngày và chỉ số UC trung bình theo từng tháng trong năm.



Q & A

